

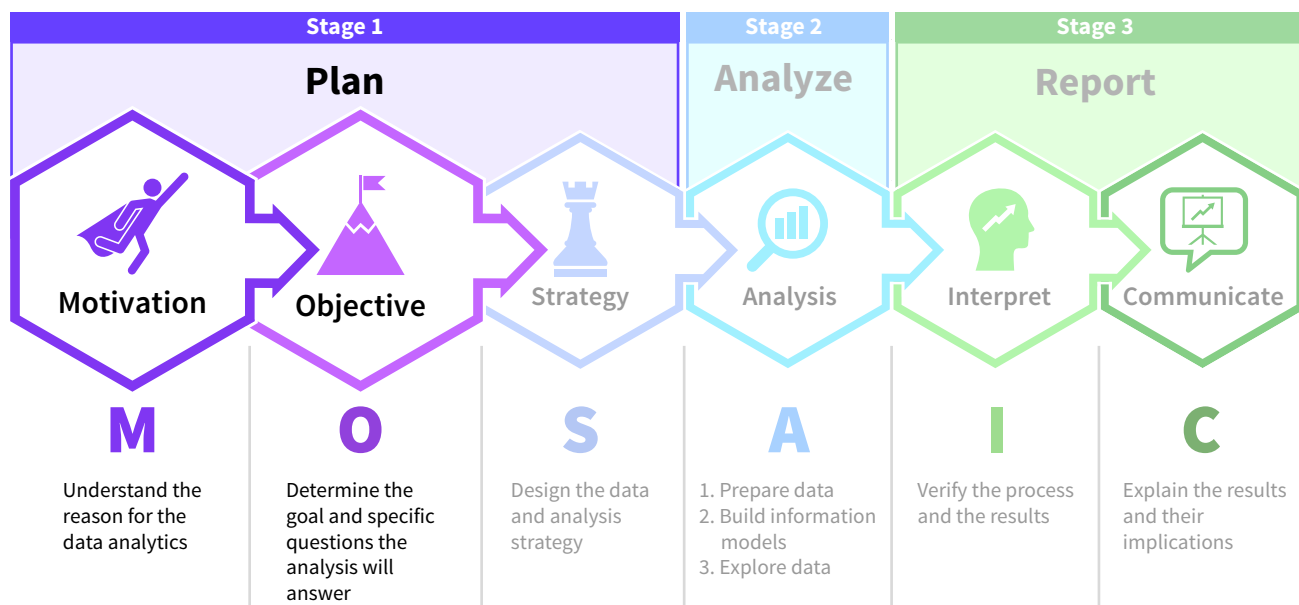
Motivations and Objectives for Data Analysis

Chapter Preview

Data analytics is a rapidly evolving and exciting field, and there is a demand within it for accounting professionals who can use new tools to analyze large amounts of data. Analyzing data and extracting information from it leads to informed decisions, and businesses today want people with these skills.

But a data analytics project does not start with a deep dive into the data. Planning the project, including understanding the motivation for performing it and developing its specific goals, is equally important. Data analytics can be costly, complex, and time consuming. A detailed plan links the project's motivation with its objective and key questions, which helps us select the best analysis method. Without a plan, it is possible to overlook important information or critical questions, which could make the analysis less effective or even useless.

The first stage of the data analytics process is planning. It includes identifying the motivation for the project, which helps determine the objective. A project's objective drives everything from the questions asked to the analysis choices we make to find the answers to them. This chapter examines the link between understanding why the project is necessary and developing a focused project objective.



Professional Insight: Why do Motivations and Objectives Matter for Data Analysis Projects?

Georgia lived with her parents while earning an undergraduate accounting degree at the local university. Immediately after graduation, she completed an auditing internship during busy season. She is entering a master's program of accounting in the fall semester.

On the first day of my internship, I learned professional accountants follow a process each time they work on a data analyses project. **First, they understand the motivation for their projects, which in turn influences their specific work objectives. When they are clear about their motivation and objectives, they can plan a successful strategy for their data analyses projects.**

I also learned that professional accountants are motivated by more than personal career goals. Auditors want to add value to their firms and clients by assuring them that their financial statements are reliable and accurate. After all, good information means better capital markets, lending decisions, government relationships, and business partnerships.

This helped me realize that I was also motivated by more than just getting that great job after college. I want to make a difference in the world, and I want to honor my parents' hard work raising me to be successful. This awareness of my motivations has helped me become more committed to my objective and my studies!

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 3.1 Summarize the relationship between motivations, objectives, and data analysis questions.	- Understanding Motivation - Clear Objectives Lead to Focused Data Analysis Questions	Link Motivation to Objectives (Example: Auditing)
LO 3.2 Demonstrate how to develop descriptive questions.	- Develop Descriptive Questions - Descriptive Analyses Examples	Describe Customers' Buying Behaviors (Example: Financial Accounting)
LO 3.3 Demonstrate how to develop diagnostic questions.	- Develop Diagnostic Questions - Diagnostic Analyses Examples	Determine the Risk of Material Misstatement of Sales (Example: Auditing)
LO 3.4 Demonstrate how to develop predictive questions.	- Develop Predictive Questions - Predictive Analyses Examples	Plan a Sales Trend Analysis (Example: Managerial Accounting)
LO 3.5 Demonstrate how to develop prescriptive questions.	- Develop Prescriptive questions - Prescriptive Analyses Examples	Prescribe Optimal Sales Mix (Example: Managerial Accounting)
LO 3.6 Describe motivations and objectives for data analytics in professional practice.	- Accounting Information Systems - Auditing - Financial Accounting - Managerial Accounting - Tax Accounting	Match Motivations to Professional Practice Areas

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

3.1 How Does Motivation Inform Objective-Based Data Analysis Questions?

LEARNING OBJECTIVE 1

Summarize the relationship between motivations, objectives, and data analysis questions.

It is easier to stay focused on a task's objective when we know why we are performing it in the first place. Understanding the motivation to perform an analysis and identifying its objective also helps create specific, objective-based questions. After all, it is impossible to get the right answers if we do not know what to ask.

Understanding Motivation

Motivation in data analytics in accounting is the reason the analysis is being performed. The “why” behind a project can vary, but data analysis projects are typically motivated by four forces:

- **Opportunity:** Evaluating new opportunities that will benefit the organization.
- **Professional issues:** Evaluating changes due to new laws, regulations, or changes in accounting practices.
- **Problem solving:** Solving a problem or issue that the organization is facing.
- **Process and performance assessment:** Understanding and improving the processes and performance of the organization.



Illustration 3.1 describes some common motivations and includes an example from a company called Super Scooters, which manufactures and sells a variety of scooters.

In each Super Scooters example, the motivation for the analysis is to either describe the situation's potential, diagnose the causal factors or determine operational change, predict sales revenues or net income, or prescribe the best course of action. Regardless of the motivation's source, the data analyses choices we make depend on what can be gained from the analysis. If its potential benefits outweigh the costs of doing it, then stakeholders will consider it a valuable analysis.

Clear Objectives Lead to Focused Data Analysis Questions

You just learned that the four common motivators for data analytics in accounting are opportunities, regulatory changes, solving problems, and process and performance evaluation. For example, a company with an underperforming business unit will evaluate its performance to understand why. A successful data analytics project depends on a more specific statement, which will then inform the questions that will be asked in the analysis.

ILLUSTRATION 3.1 Motivation for Data Analytics in Accounting

Motivation	Description	Role of Data Analysis	Super Scooters Example
Opportunities	New, potentially advantageous occasions or channels.	Reliable evidence justifies new products and services. Evidence to justify changing resource allocations and capital sourcing.	• New scooter models. • New services, such as scooter training classes. • Store expansions.
Professional Issues and Requirements	Compliance with new or upcoming regulations, laws, or changes in practices.	Understanding changes improves responses. Evaluating new accounting regulations can positively impact client financial statements.	The local government may have recently allowed scooters, unmotorized and motorized, on all paved bicycle paths and road lanes in the city.
Problem Solving	Solving problems related to risks and issues around clients, technologies, employees, operations, and supply chains.	Solving problems to minimize productivity losses.	Super Scooters may need to investigate why a particular scooter is no longer selling as well as it did, or why it is not selling as well as their other scooter products.
Process and Performance Assessment	Evaluating financial statements for material misstatements or risks of material misstatement. Inspecting operational efficiency and effectiveness at the individual, functional, unit, and business levels.	Audit data analytics can provide risk assessment and substantive audit evidence. Evaluating the success of a new strategy helps management make future decisions.	Super Scooters may need to evaluate the contribution margin and inventory turnover for each product line, or the performance of each salesperson.

Determine the Objective

The objective of a data analysis project naturally follows the motivation. Let's go back to that common motivation for data analysis, an underperforming business unit:

- The motivation is to solve a problem by evaluating the unit's performance to discover why it is underperforming.
- It is necessary to be more specific because poor performance could have several causes.
- The first step is to specify the areas of performance to investigate and articulate what those investigations should accomplish.

In other words, it is important to clarify the goal of the data analysis.

Every data analysis project begins with setting an **objective**, which is the project's goal. It is a statement that details what the project will accomplish. A new example can illustrate how to determine a project's objective. Omni Restaurants owns restaurants across five regions in the United States. The financial information for the current year by region is shown in **Illustration 3.2**.

Illustration 3.2 shows that Omni is losing money in one region:

- **Motivation:** They are concerned about the losses in Region 1 and want to understand what is driving them.
- **General objective:** Eliminate those financial losses.

Now we have moved from motivation to objective. But the objective is so general that it is difficult to decide where to begin the analysis. A specific objective would narrow the analysis' focus. In the Omni case, only one region is experiencing losses:

- **Specific objective:** Identify the factors driving the losses in Region 1.

After the objective is clear and specific, formulate the questions that will achieve that objective once they are addressed.

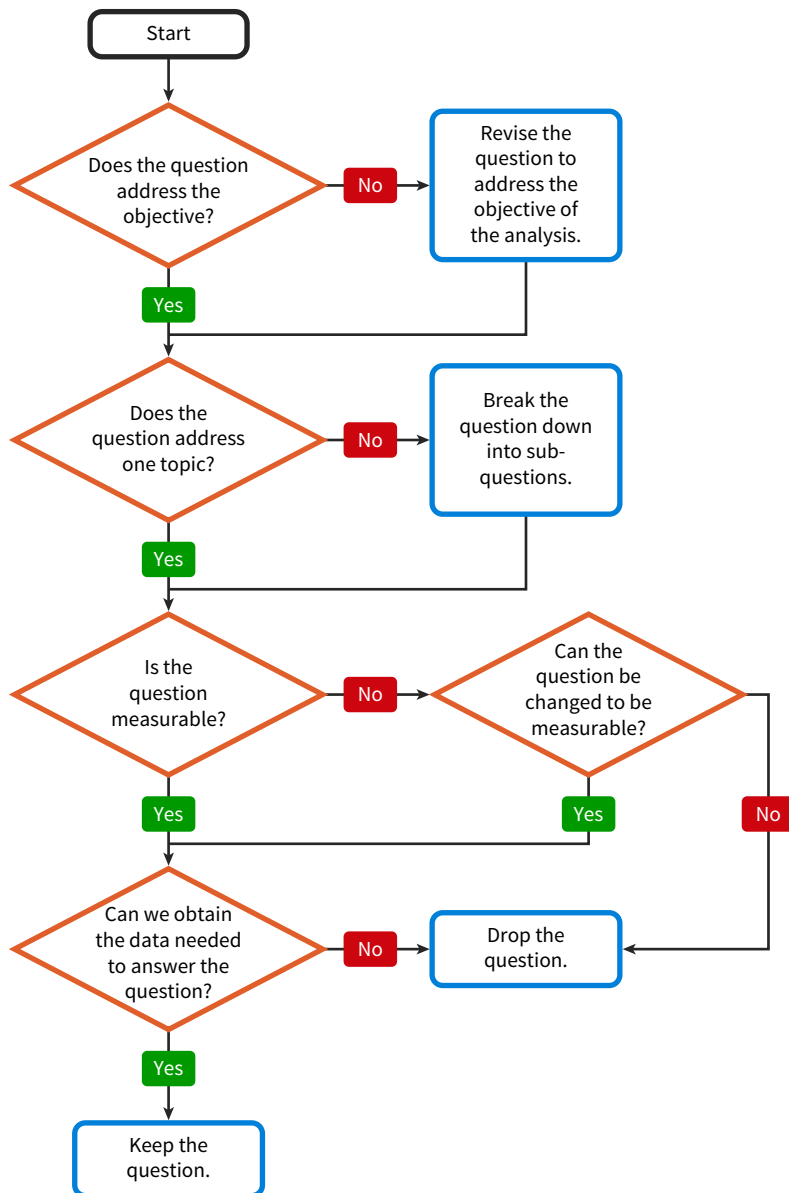
ILLUSTRATION 3.2 Omni Restaurants Financial Performance by Region

	Region 1	Region 2	Region 3	Region 4	Region 5	Total
Dine in Sales	\$226,800	\$333,396	\$367,569	\$405,245	\$446,783	\$1,779,793
Takeout Sales	\$151,200	\$222,264	\$245,046	\$270,163	\$297,855	\$1,186,528
Total Revenues	\$378,000	\$555,660	\$612,615	\$675,408	\$744,638	\$2,966,321
Cost of Goods Sold	\$ 86,940	\$122,245	\$122,523	\$128,328	\$134,035	\$ 594,071
Gross Profit	\$291,060	\$433,415	\$490,092	\$547,080	\$610,603	\$2,372,250
Lease	\$ 90,000	\$ 61,500	\$ 63,038	\$ 64,613	\$ 66,229	\$ 345,380
Marketing	\$ 30,000	\$ 25,000	\$ 25,000	\$ 25,000	\$ 25,000	\$ 130,000
Salaries	\$200,835	\$204,030	\$224,943	\$236,190	\$248,000	\$1,113,998
Other Expenses	\$ 5,250	\$ 4,000	\$ 4,500	\$ 5,000	\$ 5,500	\$ 24,250
Total S G & A Expenses	\$326,085	\$294,530	\$317,481	\$330,803	\$344,729	\$1,613,628
Depreciation	\$ 36,960	\$ 36,960	\$ 36,960	\$ 36,960	\$ 36,960	\$ 184,800
Interest	\$ 23,621	\$ 20,668	\$ 17,716	\$ 14,763	\$ 11,810	\$ 88,578
Earnings Before Tax	(\$95,606)	\$ 81,257	\$117,935	\$164,554	\$217,104	\$ 485,244
Income Tax Expenses	0	\$ 29,526	\$ 42,853	\$ 59,344	\$ 77,911	\$ 209,634
Net Income (Loss)	(\$95,606)	\$ 51,731	\$ 75,082	\$105,210	\$139,193	\$ 275,610

Articulate Questions

Asking the wrong questions leads to answers that do not address the objective. Good questions are clear, concise, and measurable. So, how are good questions developed? Evaluate each question as to whether it addresses the objective, focuses on a single issue, is measurable, and if the data necessary to answer it are available. If the question does not meet this criteria, then either revise it, break it down, or drop it. Use the flowchart in **Illustration 3.3** to develop specific data analysis questions.

ILLUSTRATION 3.3 Flowchart for Developing Data Analysis Questions

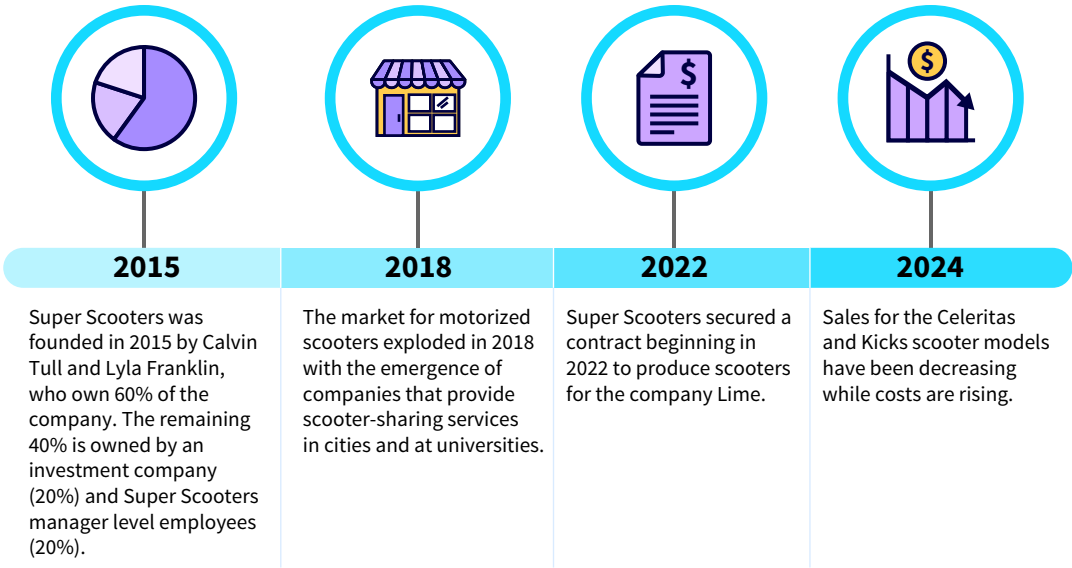


A critical thinking approach to question development can help with this process. Recall that thinking critically when performing data analytics involves considering those who are impacted by the analysis, its purpose, considering a variety of alternatives at each step, controlling for risks, acquiring the right information or knowledge, and reflecting on past and current projects.

The Role of Critical Thinking

Let’s use the Super Scooters example to show the importance of critically evaluating motivation, objectives, and questions at this stage of the data analysis process. Assume you are an accountant for the company Super Scooters. Super Scooters is a privately held company¹ that produces three models of motorized scooters (two electric and one gas-powered) and one non-motorized scooter (Illustration 3.4).

ILLUSTRATION 3.4 Super Scooters Company Timeline



Super Scooters is now in its third year of the contract with Lime, and they are unsure if they have made the right production decisions. Sales seem to be decreasing for some models, and costs are rising. Calvin and Lyla asked you to help them do three things:

- 1. Understand which models have decreasing sales and why.
- 2. Forecast future warranty expenses.
- 3. Determine the most profitable product mix for production.

Use the critical thinking framework, SPARKS, to identify the motivation, objectives, and questions relevant to the analysis Calvin and Lyla would like you to perform. Illustration 3.5 summarizes how critical thinking helps evaluate motivations, objectives, and questions.

¹A privately held company is one that does not sell their stock on public exchanges.

ILLUSTRATION 3.5 Thinking Critically About Motivations and Objectives

Critical Thinking Element		Why It Matters	Super Scooters Example
S	 Stakeholders	Considering the perspectives of stakeholders helps identify the objective and develop good questions.	The internal stakeholders are Lyla and Calvin. Their motivation is to better understand company performance. The objective is to describe current performance, diagnose issues related to poor performance, predict future sales, and make recommendations for improvements.
P	 Purpose	Critically thinking through the purpose of data analyses is the link between a defined objective and specific questions: • A purpose is the reason something is done or created. • An objective is something we plan to do, or a goal. The questions will be different depending on whether the purpose is to describe and diagnose current conditions or predict and prescribe future outcomes.	Based on the objectives, the purpose is to: • Describe what is happening now and diagnose issues related to poor performance. • Predict future sales. • Prescribe a profitable sales mix.
A	 Alternatives	Evaluating different methods of analysis increases the likelihood of selecting the best option for the project's purpose.	Develop several alternative methods for performing the analysis to describe performance over the past year. Consider how likely each option is to provide valuable information. Rank the alternatives and choose the one with the highest expected value.
R	 Risks	Being aware of potential risks makes it possible to also mitigate them when evaluating motivation, objectives, and questions.	It may be tempting to make Calvin and Lyla happy with a positive outlook in the predictive model. Be aware of that bias and develop objectives and questions that will not be influenced by it.
K	 Knowledge	Successfully completing a project requires confirming we have the necessary knowledge to do so or doing additional research to obtain that knowledge.	One purpose of the Super Scooters analysis is to predict future sales and expenses. If you do not know how to prepare a predictive analysis, you must do one of two things: • Hire someone to prepare the predictive analysis. • Learn how to do the analysis.
S	 Self-Reflection	Understanding the purpose of the analysis makes it easier to apply the experience of performing it to future data analytics contexts and tasks. Similarly, considering past experiences can help identify objectives and develop questions.	Assessing whether you should have performed the analyses may depend on whether the information lead to better decisions by management. If it did not, then consider why. Only then can you compare expected net benefits to the actual net benefits.

Apply It 3.1

Link Motivation to Objectives



Auditing Best friends Luanne and Maxine had been making baked goods for a friend who owned a local coffee shop. Their delicious baked goods soon had so many fans that Luanne and Maxine had to rent commercial kitchen space to keep up with demand. Now their business, Best Bakes Bakery, provides baked goods for a variety of coffee shops and restaurants.

They want to expand their operations into other states and are looking for potential investors. They believe a CPA's opinion on their financial statements will help convince investors to invest in their company.

The owners hired your CPA firm to prepare a review of their financial statements. Best Bakes Bakery is the first bakery business for your firm, and you are the external auditor assigned to the engagement. Your supervisor asked you to use data analytics for the risk assessment:

- You have the last three years of financial statement data and must calculate several ratios relevant to the company's operations.
- Your supervisor believes the analysis will identify areas of potentially higher risk of material misstatement.

1. Which category of motivation applies to this scenario?
2. Who are the stakeholders in your data analysis?
3. What is the objective of your analysis?

SOLUTION

1. The motivation is process and performance evaluation. The analysis of financial ratios will provide information relevant to the risk assessment of the potential for material misstatements.
2. You, your audit firm, and the client are the internal stakeholders. The client's investors and creditors are the external stakeholders.
3. The objective of the analysis is to determine the potential risk of material misstatements in the financial statements.

3.2 What are Descriptive Objectives?

LEARNING OBJECTIVE 2

Demonstrate how to develop descriptive questions.

A data analysis project with the objective to understand something that is currently happening or that has happened requires descriptive questions. These types of questions are answered using descriptive analytics. Descriptive questions are often the first asked in any analysis because understanding the data is necessary before beginning more advanced analyses.

Develop Descriptive Questions

Descriptive questions are designed to better understand data to answer business questions. To develop good descriptive questions, first identify the objective of the analysis, then break that down into questions. Remember, a good question is one that relates to the objective, is specific, measurable, and can be answered with the available data.

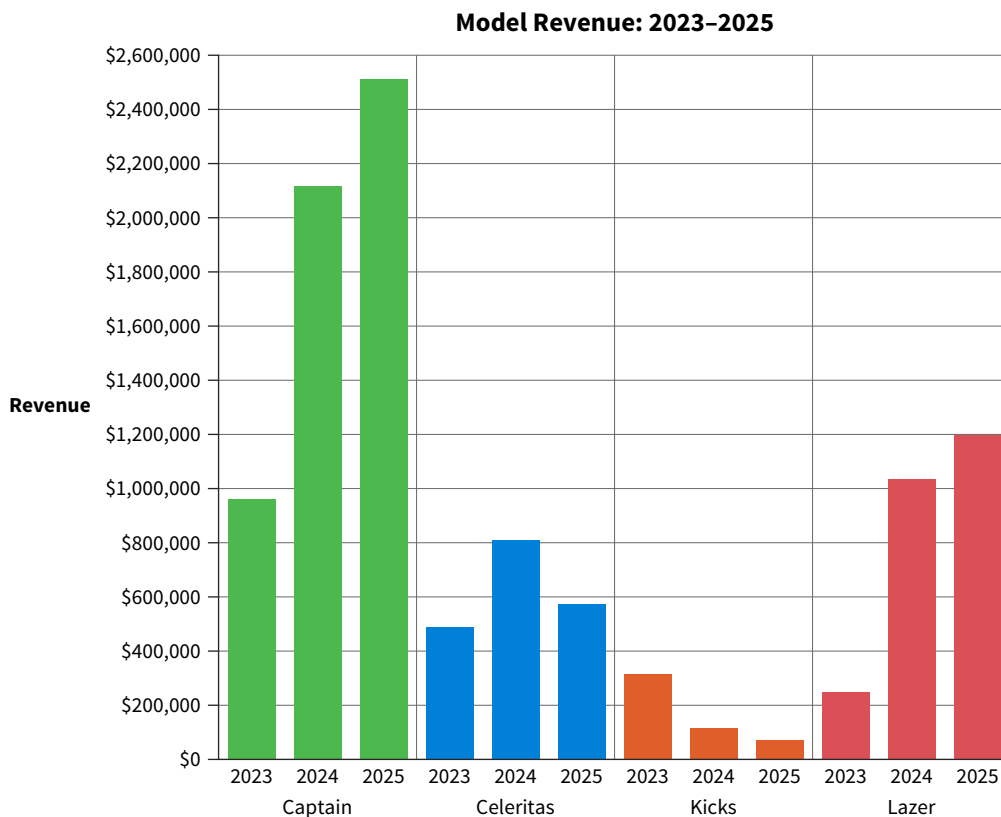
Let's continue with the Super Scooters example to illustrate. **Illustration 3.6** provides the financial information for the years 2023 through 2025.

ILLUSTRATION 3.6 Super Scooters' Income Statements: 2023–2025

	Year Ended December 31		
	2023	2024	2025
Gross Sales	\$4,924,816	\$10,506,334	\$11,378,847
Total Variable Costs	\$2,190,588	\$ 4,922,488	\$ 5,416,827
Contribution Margin	\$2,734,228	\$ 5,583,846	\$ 5,962,020
Total Fixed Costs	\$ 579,988	\$ 1,027,391	\$ 1,042,201
Net Income	\$2,154,240	\$ 4,556,455	\$ 4,919,819

While sales and net revenue have increased, two of the models have decreasing net revenue (**Illustration 3.7**).

ILLUSTRATION 3.7 Scooter Model Revenue: 2023–2025



Super Scooters wants to understand the decreases in net revenue for the Celeritas and Kicks models. What questions should they ask of the data? What possible measure(s) can be used to answer the questions?

Illustration 3.8 provides examples of both initial questions and sub-questions (in other words, more detailed questions).

ILLUSTRATION 3.8 Descriptive Questions to Understand Decreases in Revenue at Super Scooters

Objective	Initial Questions	Sub-questions	Possible Measures
Understand decreases in net revenue for the Celeritas and Kicks models.	Have sales decreased for the Celeritas and Kicks models?	1. Have sales decreased for the Celeritas model?	Gross sales dollars
		2. Have sales decreased in all locations for the Celeritas model?	Average gross sales dollars
		3. Have sales decreased for the Kicks model?	Sales volume
		4. Have sales decreased in all locations for the Kicks model?	
	Have expenses increased for the Celeritas and Kicks models?	1. Have expenses increased in all locations for the Celeritas model?	Total expense dollars
		2. Have expenses increased in all locations for the Kicks model?	Average total expense dollars Percent change from prior year
	Have variable expenses increased for the Celeritas and Kicks models?	1. Which variable expenses increased from prior year for the Celeritas model?	Variable expense dollars
		2. Do some locations have higher increases in variable expenses for the Celeritas model than others?	Average variable expenses
		3. Which variable expenses increased from prior year for the Kicks model?	Percent change from prior year
		4. Do some locations have higher increases in variable expenses for the Kicks model than others?	
	Have fixed expenses increased for the Celeritas and Kicks models?	1. Which fixed expenses increased from prior year for the Celeritas model?	Fixed expense dollars
		2. Do some locations have higher increases in fixed expenses for the Celeritas model than others?	Average fixed expenses
		3. Which fixed expenses increased from prior year for the Kicks model?	Percent change from prior year
		4. Do some locations have higher increases in fixed expenses for the Kicks model than others?	

Notice how the questions all relate to the analysis' objective. The initial questions are fine but broad. The sub-questions in the second column narrow the focus. Moreover, each question is measurable, as indicated by the measures in the last column. Once the detailed descriptive questions are outlined, it is time to determine the necessary data and analysis method to answer them.

Descriptive Analyses Examples

Analyses commonly used to answer descriptive questions include measures of frequency, location, dispersion, and percentages:

- Frequency measures help us understand categories of data.
- Measures of location (mean, median, mode) reveal average observations in a data set.
- Measures of dispersion (minimum, maximum, range, variance, and standard deviation) show how much variance there is among the observations in the data set.
- Measures of percentage change compare results to prior periods and the percent of total.

Data Let's demonstrate an example using the questions identified in Illustration 3.8.

It is possible to discover if gross sales have decreased for the Celeritas model by using the measure gross sales. Several different analyses can determine if gross sales have decreased:

- Gross sales dollars: total measure
- Sales volume: total measure
- Average unit sales price

A PivotTable in Microsoft Excel created with the Super Scooter data prepares a descriptive analysis for this question ([Illustration 3.9](#)).

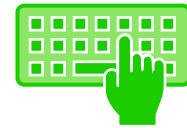
Model	Total Gross Sales	Total Sales Volume	Average of Unit Sale Price
Celeritas			
2023	\$ 1,256,478.00	3,595	\$ 350.25
2024	\$ 2,081,584.00	5,932	\$ 350.57
2025	\$ 1,488,293.00	4,258	\$ 347.92
Kicks			
2023	\$ 628,271.00	3,385	\$ 187.88
2024	\$ 279,848.00	1,816	\$ 148.37
2025	\$ 221,134.00	1,695	\$ 136.21

ILLUSTRATION 3.9 Descriptive Analysis of Celeritas and Kicks Sales

The sales data confirms the Celeritas and Kicks models each declined from 2024 to 2025. Interestingly, both sales volume and average price decreased, so the decline is a combination of both volume and price. The next step is answering the next sub-question: Did sales decrease in all locations?

[Illustration 3.10](#) shows the change in average gross sales dollars, which is a measure of location. It is the result of a descriptive analysis in Tableau that shows the change in gross sales from 2023 to 2025 by location for the Celeritas model. This is a highlight table where the darker color signals a larger decrease. ([Data](#) See [How To 3.1](#) at the end of this chapter to learn how to create this table in Tableau.)

How To



**Change in Gross Sales from
2024 – Celeritas**

	2025
Seattle	\$ (140,206)
Dallas	\$ (106,783)
Charlotte	\$ (94,794)
Boston	\$ (88,837)
Chicago	\$ (58,619)
Miami	\$ (57,786)
Salt Lake City	\$ (46,867)
Phoenix	\$ (20,787)

Difference in Gross Sales

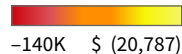


ILLUSTRATION 3.10 Descriptive Analysis of Celeritas' Change in Gross Sales by Location

The Seattle location had the largest total gross sales decrease from 2024 at \$140,206. Dallas had the second largest decrease at \$106,783. However, it is also clear that sales have dropped in all locations. The next step is to determine why sales have decreased, which will require diagnostic analytics.

Apply It 3.2

Describe Customers' Buying Behavior



Data **Financial Accounting** Best Bakes Bakery would like to better understand their top customers' buying behavior. You have been given sales transactions for the years 2022 to 2025. An excerpt from the file follows.

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930

Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Costof GoodsSold	Profit
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60
853300	Poppyseed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80

- What is the objective of the analysis?
- Develop three questions relevant to the objective, and describe the measures necessary to answer the questions.
- What analyses will you use to answer these questions?

SOLUTION

- The objective of the analysis is to identify the top customers and evaluate what products they purchase.
- | Questions | Measures |
|---|--|
| 1. Who are the top five customers? | Gross sales, sales volume, profit margin |
| 2. What are the top five products sold? | Sales volume, gross sales |
| 3. What are the spending patterns for the top five customers? | Sales volume, gross sales |
- Analyses for the three questions:
 - Descriptive analysis: Top five customers by year for each of the measures compared to average for all customers.
 - Descriptive analysis: Five highest selling products by year for each of the measures compared to average for all products.
 - Descriptive analysis: Analysis showing sales by month or quarter for each year for the top five customers. A bar chart or a line chart could be used.

3.3 What are Diagnostic Objectives?

LEARNING OBJECTIVE 3

Demonstrate how to develop diagnostic questions.

Once we know what has happened, the next step is to determine *why*. Diagnostic questions build on descriptive analyses and further explore the data to find the outcome’s cause. Diagnostic analytics do this by looking for anomalies, correlations, patterns, or trends.

Develop Diagnostic Questions

Diagnostic questions identify a problem or issue to understand why an outcome occurred. **Illustration 3.11** shows diagnostic questions based on the results of the descriptive analysis of Super Scooters’ sales in the previous section.

ILLUSTRATION 3.11 Diagnostic Questions About Super Scooters’ Sales Decline

Objective	Initial Question	Sub-questions	Possible Measures
Determine why sales are decreasing for the Celeritas and Kicks models.	What is driving the sales volume decrease?	1. Are there anomalies in sales volume for the Celeritas model? 2. Are there anomalies in sales volume for the Kicks model? 3. Are there identifiable patterns in sales volume for the Celeritas model? 4. Are there identifiable patterns in sales volume for the Kicks model?	Sales volume
Understand the large decrease in sales at the Seattle location.	What factors are driving the decrease in sales at the Seattle locations?	1. Are there unusual patterns in the Seattle location sales of Celeritas models? 2. Are there unusual patterns in the Seattle location sales of the Kicks model?	Gross sales dollars Average gross sales dollars Sales volume

Why are sales decreasing, and specifically, why did the Seattle location have the largest decrease? The questions and sub-questions dive deeper into the sales decrease by asking about anomalies and unusual patterns. Now, identify the data and analyses that will answer these questions.

Diagnostic Analyses Examples

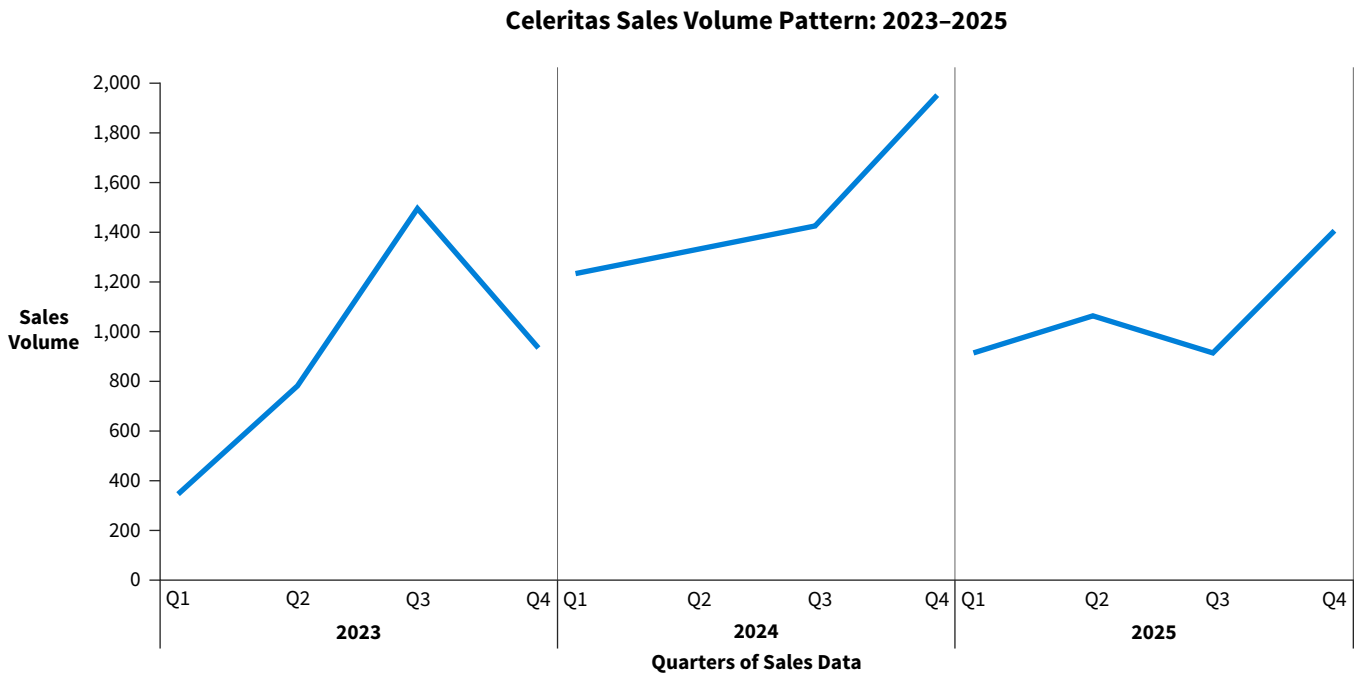
There are four common types of diagnostic analyses: anomaly detection, correlation, pattern detection, and trend analysis.

Using the Super Scooters questions in Illustration 3.11, let’s examine these sub-questions:

- 1. Are there identifiable patterns in sales volume for the Celeritas model?
- 2. Are there unusual patterns in the Seattle location’s Celeritas model sales?

Line charts can identify patterns by revealing repeating patterns in the data. [Illustration 3.12](#) shows Celeritas sales by month for each year.

ILLUSTRATION 3.12 Celeritas Sales Volume Patterns



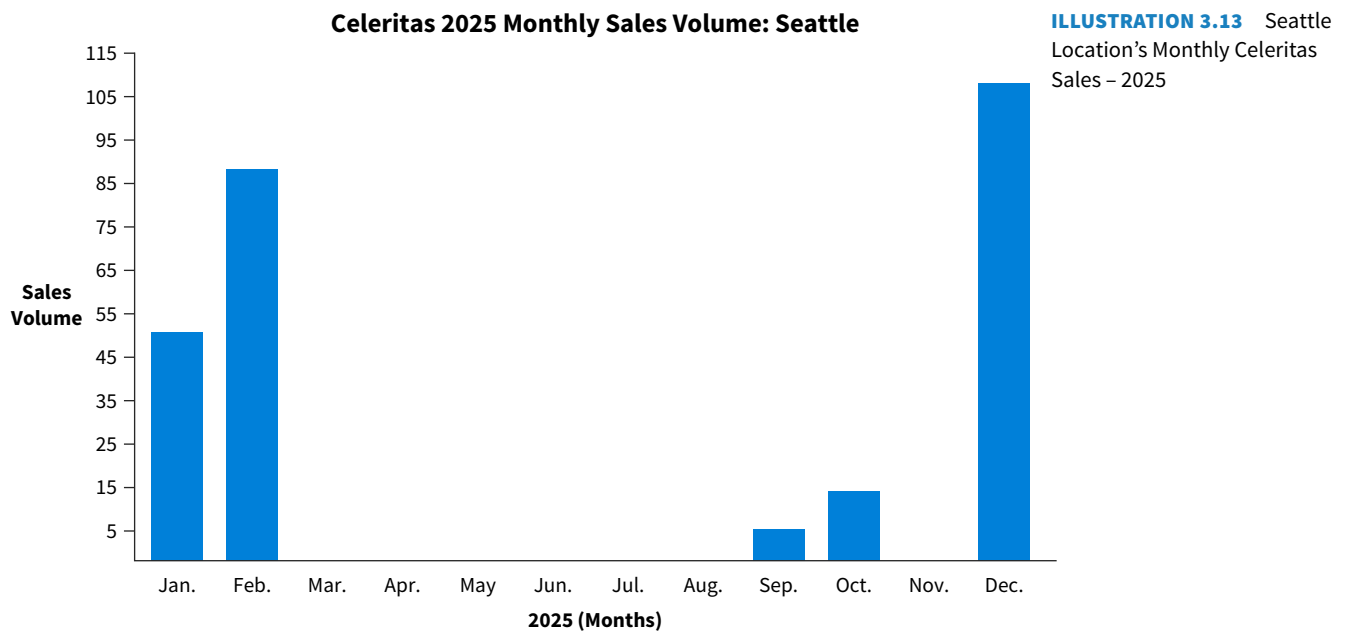
The line charts in Illustration 3.12 do not appear to show a pattern in sales volume for the Celeritas model. However, the line chart shows all locations, and previous analyses revealed the Seattle location had a drop in Celeritas sales.

The sales pattern in years 2023 and 2025 indicate that sales generally increase from the first quarter to the third quarter:

- In 2023 there is a drop in sales in the fourth quarter, but 2024 and 2025 saw an increase in fourth quarter sales.
- However, 2025 sales are below 2024 sales, and there was a large drop in the third quarter.

To understand why sales are lower in 2025 and why there was a drop in the the third quarter of 2025, we must further analyze sales volume by location.

[Illustration 3.13](#) shows an analysis examining monthly sales of the Celeritas at the Seattle location.



The column chart of Celeritas sales over 2025 shows one thing and suggests another:

- In 2025 there are several months with no sales.
- This could be a contributing factor to the decline in overall Celeritas sales in 2025.

The next step would be asking management what happened in March 2025 that led to six months without any sales of the Celeritas model. This also raises additional questions to investigate:

- Did sales drop at the Seattle location only?
- Was the Celeritas model the only product with declining sales?

This investigative process is called exploratory data analysis, which you learn about later in this course.

Data Auditing During the Best Bakes Bakery financial statement audit, you have been asked to determine if there are unusual changes in sales from prior years that might affect risks of material misstatement. You have been given an excerpt of the transactions in an Excel file so that you know the data available.

Apply It 3.3

Determine the Risk of Material Misstatement of Sales

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930



Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Costof GoodsSold	Profit	Profit Margin
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60	0.463768
853300	Poppyseed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40	0.587629
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00	0.529412
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50	0.359375
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60	0.646341
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50	0.160193
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80	0.333333

1. What is the objective of the analysis?
2. Develop three questions relevant to the objective, and state the measures necessary to answer them.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective is to analyze the transactions from 2022 to 2025 for unusual changes. The analysis will provide the auditor with information to determine the nature, timing, and extent of audit procedures.

2. Questions	Measures
1. Are there any unusual changes in total revenue over the years 2022–2025?	Quarterly gross sales
2. Are there any unusual changes in total revenue over the years by location?	Quarterly gross sales
3. Are there any unusual changes in revenue by product over the years?	Sales volume by product

3. Anomaly, pattern, and trend analyses can be prepared to answer the questions.

3.4 What are Predictive Objectives?

LEARNING OBJECTIVE

4

Demonstrate how to develop predictive questions.

So far, you have learned how to develop descriptive questions to find out what happened in the past and diagnostic questions to understand why. What if you want to know what may happen in the future? In this case, you would ask **predictive questions**.

Develop Predictive Questions

When making loan decisions, banks use historical information about whether a potential borrower pays bills consistently and on time to predict if they will continue to do so. In the same way, predictive analytics use past and present data to create models so businesses can make predictions.

The use of predictive analytics is not new in the accounting profession, but as the availability of data and software tools to perform predictive analytics has increased, so has performing these analytics in all areas of accounting:

- Financial accountants can identify trends in sales or expenses.
- Cost accountants can predict costs, create forecasts, and evaluate cost drivers.
- Auditors identify potential material misstatements using predictive analytics.
- Tax accountants may use predictive analytics for tax planning.

Suppose Super Scooters is preparing their budget for the next year:

- They want to predict revenue for 2026 assuming a 10% increase in sales volume.
- They also believe there will be a 10% increase in warranty costs.
- Finally, they are considering discontinuing the Celeritas model and want to know if that will change predicted revenue.

What is the overall objective of the data analysis? What questions should Super Scooters ask to meet that objective?

When determining predictive questions, it is helpful to ask, “What do I want to do with the answer?” **Illustration 3.14** shows the overall objective of predicting revenue for 2026, the initial question, plus specific sub-questions.

Objective	Initial Question	Sub-questions
Predict revenue for 2026.	How much will revenue change with a 10% increase in sales volume?	How much will revenue change for each model? How much will revenue change at each location?
	How will revenue be affected with a 10% increase in warranty costs?	What factors influence warranty expense?
	How will revenue be affected by discontinuing the Celeritas and/or the Kicks model?	How much will revenue at each location be affected by discontinuing the Celeritas and/or the Kicks model?

ILLUSTRATION 3.14 Predictive Questions to Forecast Super Scooters' Revenue

Like all questions, predictive questions should be related to the objective, specific, measurable, and the appropriate data and analytics should be used to answer them.

Predictive Analyses Examples

Two common analyses that answer predictive questions are trendlines and regression analysis.²

Trendlines

Trendlines show the underlying functional relationship of data:

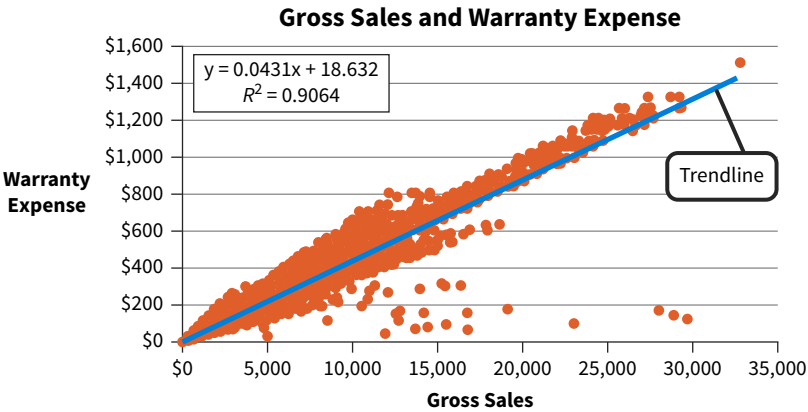
- A **functional relationship** is the effect of an independent variable on a dependent variable.
- A **linear function** shows steady increases or decreases over the range of the independent variable.

The Excel Trendline tool can help determine if the data follow a linear function. Remember that Super Scooters believes warranty costs will rise by 10%. Further, warranty costs are believed to be driven by sales.

²There are many other predictive analytical methods, such as artificial intelligence, which are beyond the scope of this chapter.

Illustration 3.15 shows a trendline created in Excel to determine if there is a linear relationship between sales and warranty expense for Super Scooters.

ILLUSTRATION 3.15 Trendline
Using Sales and Warranty Expense



The trendline supports a linear relationship:

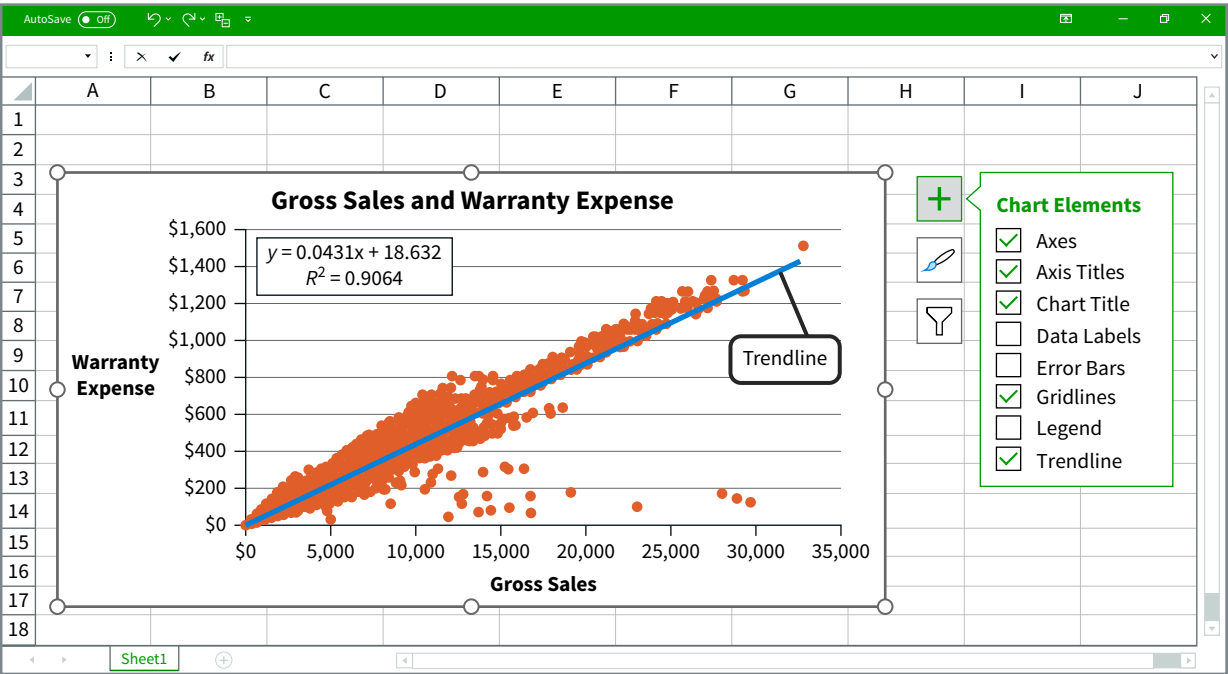
- As gross sale amounts increase, warranty expense also increases.
- The equation for the trendline is shown in the chart. ($y = 0.0431x + 18.632$, $R^2 = 0.9064$).

That equation can be used to predict future warranty expense based on expected sales where y is the amount of warranty expense and x is the amount of the gross sale. For example, if a gross sale was \$2,000, then the expected warranty expense would be:

$$(0.0431 \times 2,000) + 18.632 = \$104.83$$

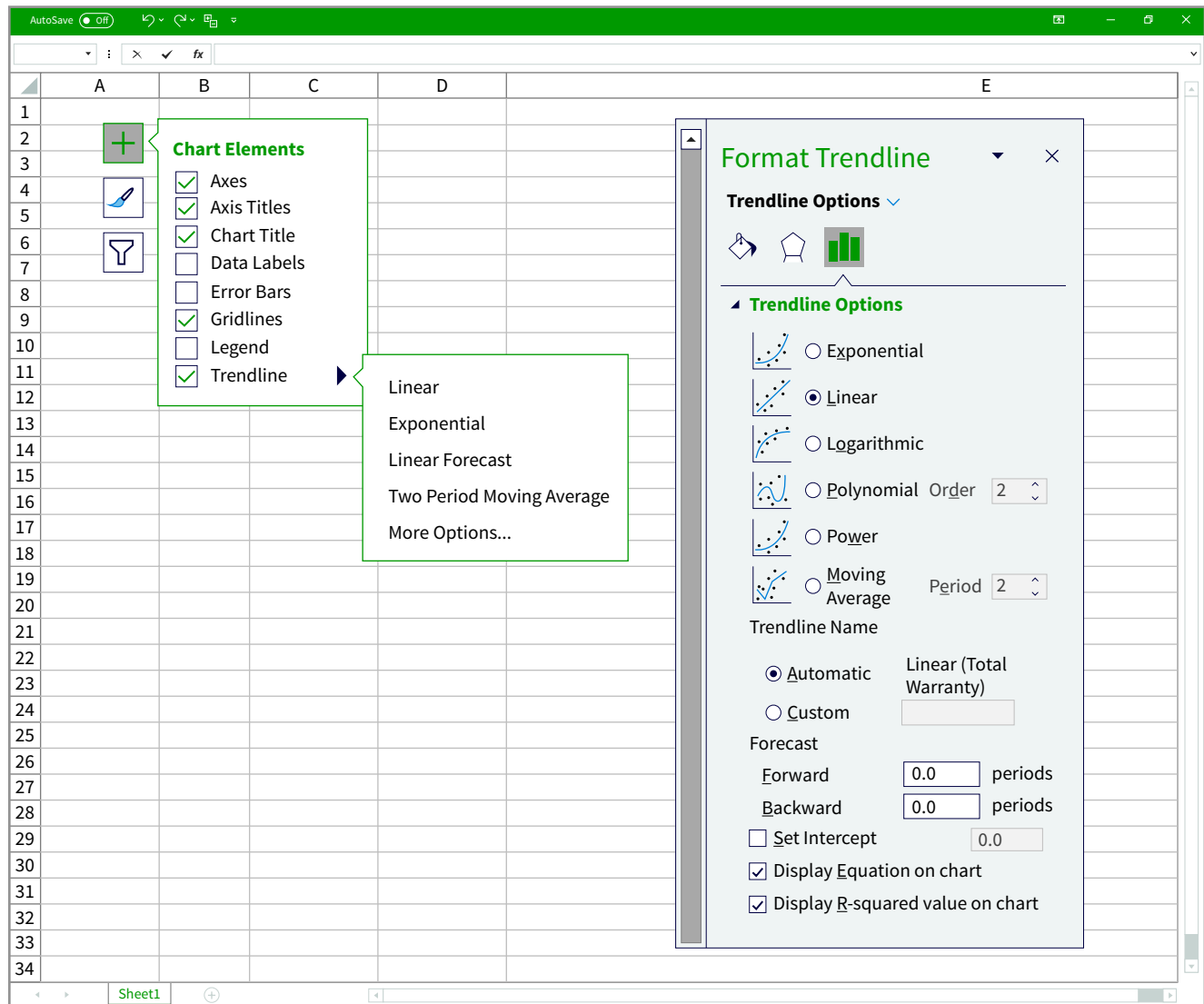
If warranty costs increase by 10% as Super Scooters expects, the prediction model can be adjusted by increasing 0.0431 by 10%. Find the trendline tool **Chart Elements** in Excel by clicking on the graph and then the plus sign (**Illustration 3.16**).

ILLUSTRATION 3.16 Trendline Choice in Chart Elements



Note that clicking on the plus sign at the top left of the chart includes the option to select **Trendline**, and then **More Options**. (Illustration 3.17).

ILLUSTRATION 3.17 Trendline Options in Microsoft Excel



The **Format Trendline** box that opens lets the user choose other functions (exponential, logarithmic, polynomial, power, and moving average). This can be useful if the pattern of the data does not appear linear. Also, click on the boxes to display the equation and the *R*-square on the chart. The *R*-square is a measure of how well the line fits the data. The closer to 1, the better the fit. The trendline in Illustration 3.16 has a very strong *R*-square of 0.9064. We will discuss *R*-square in more detail in the next section.

Linear Regression

While you may not prepare a predictive model during your career, you will likely use predictive models or need to interpret their output. **Linear regression** is a tool for building mathematical and statistical models to explain the relationship between a dependent variable and one or more independent variables.

Predictive analytics build models to predict or better understand a phenomenon. To find out what factors influence warranty expense, we would build a model that predicts warranty expense.

Building a model requires identifying the variables that will be included in it:

- A **variable** is a data field used for analysis.
- A **dependent variable** is the outcome measure (warranty expense).
- **Independent variables** are the variables that influence the dependent variable (specific variables that we believe influence warranty expense, such as sales or number of warranty claims).

Simple linear regression involves a single independent variable, while multiple regression involves two or more independent variables. The goal of a regression model is finding the equation of the line that best fits the data.

Let's examine the output of a regression model to understand how they help answer questions. As an accountant at Super Scooters, you are trying to estimate next year's maintenance expense for the machinery equipment. You believe the cost drivers for maintenance expense are the number of hours the equipment is used and the number of repair requests. The regression model variables would be:

- **Dependent variable:** Maintenance expense
- **Independent variables:** Machine hours and repair requests

Historical data from the previous 36 months will be used to create a regression model. The output from the model is shown in **Illustration 3.18**.

ILLUSTRATION 3.18 Regression Model Output

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.97868787				
R Square	0.95782995				
Adjusted R Square	0.95527419				
Standard Error	1283.82134				
Observations	36				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	5252.85845	897.8427419	5.850533	1.5E-06	
Machine Hours	3.56920298	0.43769194	8.154601	2.05E-09	
Repair Requests	759.836675	168.7763028	4.502034	7.94E-05	

The regression in Illustration 3.18 was performed using Microsoft Excel. The summary output is split into three sections.

Regression Statistics First are the **regression statistics**, or the statistical measures used to evaluate the model. **Illustration 3.19** shows the regression statistics from Illustration 3.18, along with the definition of the statistic and an interpretation for the Super Scooters regression model.

ILLUSTRATION 3.19 Regression Statistics

Regression Statistics	Definition	Output	Interpretation
Multiple R	- Also called the correlation coefficient, it measures the strength of the relationship between the dependent and independent variables. - It is a measure from -1 to 1. A positive number equals a positive correlation, so the variables move in the same direction. A negative number equals a negative correlation, where the variables move in opposite directions.	0.97868787	- Multiple R is 0.978. This is a positive correlation between total expenses and the independent variables in the model (machine hours, repair requests). - As the variables in the model increase, total maintenance expenses will also increase.
R Square (R^2)	- Also called the coefficient of determination, it is a measure of how well the regression line fits the data. R^2 gives the proportion of the variation in the dependent variable that is explained by the independent variables. - The closer the R^2 is to 1, the better the regression line fits to the data.	0.95782995	This shows that 95.8% (0.9578) of the change in total maintenance expense is explained by how many hours the equipment is used (machine hours) and how many repair requests are made.
Adjusted R^2	- Explains how well the regression line fits the data. - Modifies the value of R^2 by incorporating the sample size and the number of independent variables. - Generally used to evaluate a multiple regression model. - The closer the adjusted R^2 is to 1, the better the fit of the regression line to the data.	0.95527419	Adjusted R^2 is interpreted in the same way as R^2 . This shows that 95.5% of total expenses can be explained by how many hours the equipment is used and the number of repair requests made.
Standard Error	- Represents the variability of the observed dependent variable values from the values predicted by the model. - If the data are clustered close to the regression line, then the standard error is small (optimal). - If the data are more scattered, then it is larger.	1,283.82134	- The standard error is \$1,283.82. - To determine if standard error is large or small, compare it to the standard deviation of the dependent variable. - In this example, compare the standard error to the standard deviation of total expenses. If the standard error is smaller than the standard deviation, the standard error is acceptable. - The mean and standard deviation for total expenses is: - Mean: \$27,450 - Standard deviation: \$6,070
Observations	The number of observations included in the data set.	36	In the Super Scooters example, it is 36 months of data.

Analysis of Variance (ANOVA) The next section of the regression output is the ANOVA (analysis of variance). **Illustration 3.20** is the ANOVA section from the regression model.

ILLUSTRATION 3.20 ANOVA Regression Statistics

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			

In a multiple linear regression such as this, significance is a hypothesis test of whether the regression model is better than a model with no independent variables. In other words, is the model better than no model at all?

Generally, a model is considered significant if the F statistic (*Significance F* in Illustration 3.20) is less than 0.05:

- If F is significant, then the model can explain some of the variation of the dependent variable. In other words, it is better than no model at all.
- The ANOVA in Illustration 3.20 has a significance F of 2.05385E-23. The notation “E-23” after 2.05385 represents scientific notation, also known as exponential notation. 2.05385E-23 is the same as 0.0000000000000000000000205385. It is clearly a number well below 0.05, so the model is significant.

In other words, the independent variables can explain some of the variation of total expenses, so the model is better than no model at all.

Regression Equation The last section of the regression summary output provides the information to create the equation to predict the dependent variable:

- The intercept and coefficients of the model represent the equation of the line that best fits the data.
- The key statistic to analyze in this section is the p -value for each of the independent variables. Like the F statistic, the p -value provides a test of significance. In the case of p -value, it is a test as to whether the independent variable improves the ability of the model to better predict the dependent variable. A p -value of 0.05 or less is generally considered significant.

Let’s use the output in [Illustration 3.21](#) to identify the prediction model for machine maintenance costs and interpret the coefficients.

ILLUSTRATION 3.21 Regression Model

	Coefficients	Standard Error	t Stat	P -value	
Intercept	5252.85845	897.8427419	5.850533	1.5E-06	
Machine Hours	3.56920298	0.43769194	8.154601	2.05E-09	
Repair Requests	759.836675	168.7763028	4.502034	7.94E-05	

Notice that the p -values for the independent variables meet the test of being less than 0.05 and are therefore significant. The prediction model will be equal to the intercept plus (or minus if the number is negative) the coefficients of the independent variables times the predicted values for those variables. Based on the regression model in Illustration 3.21, the equation to predict total purchasing department costs is:

$$\$5,252.86 + \$3.57 (\text{number of Machine Hours}) + \$759.84 (\text{number of repair requests})$$

The calculation of predicted total expenses if there are 2,250 machine hours in one month and 8 repair requests is shown in [Illustration 3.22](#). Start with the intercept and add the product of each independent variable coefficient and variable predicted value for a prediction of \$19,364.08 of total maintenance costs for the year.

ILLUSTRATION 3.22 Prediction Model Example

	Model Coefficient	Variable Values	Prediction
Intercept	\$ 5,252.86	1	\$ 5,252.86
Machine Hours	\$ 3.57	2250	\$ 8,032.50
Repair Requests	\$ 759.84	8	\$ 6,078.72
			\$ 19,364.08

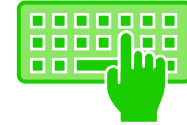
The model could be interpreted as follows:

- **Intercept:** The intercept does not always have a practical interpretation. It is a result of the model that represents the mean for the response when all the independent variables are zero. It is where the equation function crosses the y-axis. However, here the intercept represents the amount of fixed costs that exist regardless of machine hours and repair requests
- **Machine hours:** Each hour machine equipment is used adds \$3.57 to total costs.
- **Repair requests:** Each request adds \$759.84 to total costs.

Using a model like the one in Illustration 3.22 helps businesses predict future outcomes.

(Data) See How To 3.2 to learn how to perform this regression in in Microsoft Excel.)

How To



Data Managerial Accounting As a managerial accountant for Best Bakes Bakery, you are preparing an analysis of sales trends to help create the 2026 operating budget. You have been given sales transactions for the years 2022–2025. Following is an excerpt from the file.

Apply It 3.4 Plan a Sales Trend Analysis

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930



Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Costof GoodsSold	Profit	Profit Margin
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60	0.463768
853300	Poppysed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40	0.587629
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00	0.529412
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50	0.359375
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60	0.646341
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50	0.160193
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80	0.333333

1. What is the objective of the analysis?
2. Develop three questions relevant to the objective, and state the measures necessary to answer them.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective of the analysis is to predict sales for the next year's operating budget.

Questions	Measures
How are sales trending from 2022 to 2025?	Sales volume, average sales price
How are sales trending by product from 2022 to 2025?	Sales volume, average sales price
How are sales trending by location from 2022 to 2025?	Sales volume, average sales price

3. Trendline analysis will provide an estimate of sales trends that can then be applied to the budget for 2026.

3.5 What are Prescriptive Objectives?

LEARNING OBJECTIVE 5

Demonstrate how to develop prescriptive questions.

You have learned about questions that help describe what happened, investigate why, and forecast what will happen next. Next, let’s examine the questions we ask when we want to know what *should* happen.

Develop Prescriptive Questions

Prescriptive objectives build on descriptions of the present and predictions about the future to determine the best action. **Prescriptive questions** investigate how to take advantage of future opportunities or mitigate a future risk outcome. The analyses specify the actions necessary to achieve the desired outcomes.

In the Super Scooters example, we answered the previous predictive question regarding warranty expenses for 2026. Now, we can address the third analysis requested by Super Scooters—deciding how many units of each model should be produced to meet the 2026 revenue goal. The first step is to clearly articulate the questions.

Illustration 3.23 starts with the objective of determining the most profitable product mix:

- The initial question is more specific. How many scooters should be produced and sold to maximize contribution margin?
- Next, dig deeper by asking about potential constraints that should be considered and how many units of each model should be produced.

ILLUSTRATION 3.23
Prescriptive Question to Determine Super Scooters’ Product Mix

Objective	Initial Question	Sub-questions
Identify the most profitable product mix.	How many scooters should we produce and sell to maximize contribution margin?	Are there constraints that should be included in the optimization model? How many units of each model scooter should be produced to maximize contribution margin and satisfy any constraints?

Prescriptive analyses are performed with specific analysis methods and data.

Prescriptive Analyses Examples

The two most common methods of analysis used to answer prescriptive questions are optimization models and what-if analyses.

Linear Optimization

Optimization is the process of selecting values of variables that minimize or maximize some quantity of interest. Optimization modeling helps managers allocate resources more

efficiently and make cost/profit decisions. The most common optimization model used in accounting is **linear optimization**. In linear optimization, the model is comprised of:

- **Decision variables:** The unknown values the model seeks to determine.
- **Objective function:** The mathematical equation that describes the output target to minimize or maximize.
- **Constraints:** The limitations, requirements, or other restrictions that must be imposed on any solution, such as demand, material, or labor constraints.

The output from a linear optimization model will show the optimal solution.

Super Scooters' management has decided to continue producing both the Celeritas and the Kicks models for at least one more year. They want to know how many of each model should be produced to maximize contribution margin. They have forecasted demand for each model:

- Captain: 18,000 units.
- Celeritas: 10,000 units.
- Kicks: 7,000 units.
- Lazer: 24,000 units.

Because they want to avoid excess inventory, they do not want to produce more than they expect to sell. There is also a limit to the number of machine hours available for the year.

Illustration 3.24 shows information needed to create an optimization model for Super Scooters.

ILLUSTRATION 3.24 Contribution Margin and Resource Requirements

	Captain	Celeritas	Kicks	Lazer	Total
Number Produced	1	1	1	1	
CM per Unit	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	
Total CM	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	\$ 840.10
Machine Hours Per Unit	20	25	10	18	
Total Machine Hours	20	25	10	18	73
Constraints:					
Machine Hours					1,000,000
Demand	18,000	10,000	7,000	24,000	

The number of units produced are the decision variables.

The objective function is the formula for total contribution margin. Contribution margin is sales minus variable expenses.

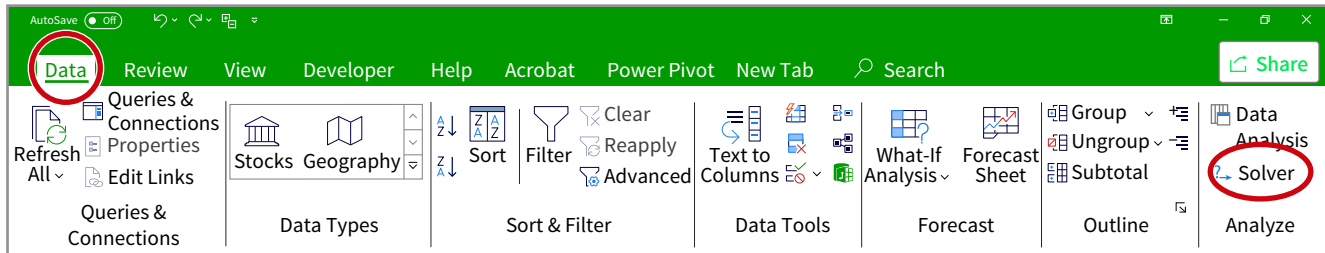
The model also shows how many machine hours are required to produce each unit.

The model shows the constraints of predicted demand for each scooter and the total machine hours available for the year.

The linear optimization program will use this data to solve for the optimal number of each model that should be produced where contribution margin (objective function) is maximized subject to the constraints. Note that we started with an arbitrary number of one in the number of units produced cells. This could also be zero to start; however, using the number one makes it possible to confirm the formulas. The linear optimization available in Microsoft

Excel Solver can illustrate how optimization models work. The Solver program is accessed via the **Data** tab in the toolbar ribbon ([Illustration 3.25](#)).

ILLUSTRATION 3.25 Accessing Microsoft Excel Solver



Clicking on **Solver** opens a dialog box to input the objective function cell, decision variable cell, and create any relevant constraints. [Illustration 3.26](#) is the Solver dialog box used to create the optimization program for Super Scooters.

ILLUSTRATION 3.26 Solver Dialog Box Inputs for Super Scooters Contribution Margin Optimization

 A screenshot of the 'Solver Parameters' dialog box. The 'Set Objective:' field is set to '\$M\$6', with a callout box labeled 'Total CM' pointing to it. The 'To:' section has 'Max' selected. The 'By Changing Variable Cells:' field is set to '\$I\$4:\$L\$4', with a callout box labeled 'Cell range for Number Produced' pointing to it. The 'Subject to the Constraints:' list contains five constraints: '\$I\$4 <= 18000', '\$J\$4 <= 10000', '\$K\$4 <= 7000', '\$L\$4 <= 24000', and '\$M\$9 <= 1000000'. A callout box labeled 'Constraints for demand and machine hours' points to this list. The 'Make Unconstrained Variables Non-Negative' checkbox is checked. The 'Select a Solving Method:' dropdown is set to 'Simplex LP'. At the bottom, there are 'Help', 'Solve', and 'Close' buttons.

Once all the constraints are input, check the box to ensure the Solver results are not negative (we cannot “unproduce” a product) and select the **Solving Method Simplex LP**, since this is a linear optimization. Clicking **Solve** produces the box in [Illustration 3.27](#).

ILLUSTRATION 3.27 Solver Results Choice Box

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

☐ Return to Solver Parameters Dialog
 ☐ Outline Reports

Reports

Answer
Sensitivity
Limits

OK Cancel Save Scenario...

Solver found a solution. All Constraints and optimality conditions are satisfied.

When the GRG engine is used, Solver has found at least a local optimal solution. When Simplex LP is used, this means Solver has found a global optimal solution.

This shows that Solver found an optimal solution that satisfied the constraints. The default is to **Keep Solver Solution**. If this radio button is checked, the spreadsheet will reflect the new decision variable amounts and the optimal contribution margin amount ([Illustration 3.28](#)). There is also the choice to create three reports. Choose the **Answer** report and click **OK**.

ILLUSTRATION 3.28 Optimization Model after Running Solver

	Captain	Celeritas	Kicks	Lazer	Total
Number Produced	18,000	5,520	7,000	24,000	
CM per Unit	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	
Total CM	\$ 6,692,106.26	\$ 1,114,328.39	\$ 729,872.21	\$ 3,892,219.65	\$ 12,428,526.51
Machine Hours Per Unit	20	25	10	18	
Total Machine Hours	360,000	138,000	70,000	432,000	1,000,000
Constraints:					
Machine Hours					1,000,000
Demand	18,000	10,000	7,000	24,000	

The number of units produced are the decision variables.

The objective function is the formula for total contribution margin. Contribution margin is sales minus variable expenses.

The model also shows how many machine hours are required to produce each unit.

The model shows the constraints of predicted demand for each scooter and the total machine hours available for the year.

Illustration 3.29 is the answer report. The first section of the report shows the objective function's original value and then the final value when the optimal solution is reached. In this case, the optimal production will be 18,000 Captain, 5,520 Celeritas, 7,000 Kicks, and 24,000 Lazer scooters.

ILLUSTRATION 3.29 Solver Results Output

Objective Cell (Max)			
Cell	Name	Original Value	Final Value
\$M\$6	Total CM Total	\$ 840.10	\$ 12,428,526.51

Variable Cells				
Cell	Name	Original Value	Final Value	Integer
\$I\$4	Number produced Captain	1	18,000	Contin
\$J\$4	Number produced Celeritas	1	5,520	Contin
\$K\$4	Number produced Kicks	1	7,000	Contin
\$L\$4	Number produced Lazer	1	24,000	Contin

Constraints					
Cell	Name	Cell Value	Formula	Status	Slack
\$M\$9	Total Machine Hours Total	1,000,000	\$M\$9<=\$M\$12	Binding	0
\$I\$4	Number produced Captain	18,000	\$I\$4<=\$I\$13	Binding	0
\$J\$4	Number produced Celeritas	5,520	\$J\$4<=\$J\$13	Not Binding	4,480
\$K\$4	Number produced Kicks	7,000	\$K\$4<=\$K\$13	Binding	0
\$L\$4	Number produced Lazer	24,000	\$L\$4<=\$L\$13	Binding	0

The middle section of the report (Variable Cells) shows the final value for the decision variables (**Illustration 3.30**). It shows the number of each model scooter that Super Scooters should sell to achieve maximum contribution margin.

ILLUSTRATION 3.30 Solver Results Output – Variable Cells Report

Variable Cells				
Cell	Name	Original Value	Final Value	Integer
\$I\$4	Number produced Captain	1	18,000	Contin
\$J\$4	Number produced Celeritas	1	5,520	Contin
\$K\$4	Number produced Kicks	1	7,000	Contin
\$L\$4	Number produced Lazer	1	24,000	Contin

Finally, the last section of the report shows how much of the constraints were used in the optimal solution (**Illustration 3.31**). The Status column indicates whether the constraint is binding or not binding. In other words, additional production is not possible without an increase in the constraint. The amount shown in the Slack column represents the amount of the constraint that is remaining after the optimal solution.

ILLUSTRATION 3.31 Solver Results Output – Constraints Report

Constraints				
Name	Cell Value	Formula	Status	Slack
Total Machine Hours Total	1,000,000	\$M\$9<=\$M\$12	Binding	0
Number produced Captain	18,000	\$I\$4<=\$I\$13	Binding	0
Number produced Celeritas	5,520	\$J\$4<=\$J\$13	Not Binding	4,480
Number produced Kicks	7,000	\$K\$4<=\$K\$13	Binding	0
Number produced Lazer	24,000	\$L\$4<=\$L\$13	Binding	0

Since machine hours are limited, the optimal model prescribes the best use of those hours to maximize contribution margin is to produce all Super Scooters can sell of the Captain, Kicks, and Lazer models and 4,480 less than demand of the Celeritas model. Any other mix will result in less contribution margin than the optimal model.

What-if Analyses

A spreadsheet model that evaluates changes and specific combinations of model inputs and assumptions is called a **what-if analysis**. A what-if analysis is an easy way to change values in a spreadsheet and recalculate the outputs. Microsoft Excel has three tools built into the Data tab under What-if Analyses. Two of these tools – Scenario Manager and Goal Seek – are

useful tools to facilitate what-if analyses. We will discuss each tool in a later chapter, but here is a short explanation:

- Scenario Manager in Excel allows changing or substituting input values for multiple cells (maximum 32). Therefore, the results of different input values or scenarios can be viewed at the same time.
- Goal Seek is used when the desired result is already known but the input value to achieve that result is not. Goal Seek is limited because it can only use one input variable. If the analysis being performed requires more than one variable to change, then an optimization model using Excel Solver is necessary. For example, the Super Scooters optimization model had more than one variable because it was necessary to consider the demand and machine hours constraints.

Data Managerial Accounting You are a managerial accountant for Best Bakes Bakery asked to prepare an analysis to determine the optimal mix of products to maximize profit. You have been given sales transactions for the years 2022–2025. Besides prior sales data, you know there are some resource constraints (such as supplies, or labor hours) that should be included in the analysis.

1. What is the objective of the analysis?
2. Develop three questions that will be relevant to the objective.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective is to determine the optimal sales mix of products given the resources available.
2. Three questions:
 - What resource constraints should be included in the decision?
 - What are the resource requirements for each product?
 - What is the expected profit per product?
3. Linear optimization can be used to determine the best mix of products for maximum profit.

Apply It 3.5 Prescribe Optimal Sales Mix



3.6 What are Data Analytics Motivations and Objectives in Professional Practice?

LEARNING OBJECTIVE 6

Describe motivations and objectives for data analytics in professional practice.

While the methods of analysis – descriptive, diagnostic, predictive, and prescriptive – are the same across accounting areas, project objectives and what motivates them can be different given the variety of purposes and stakeholders. In auditing, stakeholders are largely external (e.g. shareholders, regulatory agencies), whereas stakeholders in managerial accounting are primarily internal (e.g. management, employees). Their perspectives help identify the objective and develop good questions.

Accounting Information Systems

Motivations for data analyses in AIS can range widely in scope. **Illustration 3.32** provides a summary of typical AIS data analysis motivations, objectives, and questions.

ILLUSTRATION 3.32 Motivations, Objectives, and Questions for AIS Data Analytics

Motivation	Objective	Question
Investing in new technologies.	Determine the technology with the highest return on investment.	What are the labor cost savings if we adopt a new technology?
Increasing security of AIS assets, including data.	Determine how many security breaches or attempted breaches have occurred.	What is the total number of attempted security breaches per day?
Innovations in operational processes, either human or computer, to improve processes.	Identify inefficient processes.	Which processes take the longest?
Improving system performance, such as processing time, reliability, and availability.	Evaluate processing time for monthly financial statement reporting.	How many hours does it take to process financial statements each month?
Improvements in AIS system maintenance.	Evaluate how long it takes to address system maintenance issues.	How long does it take to resolve maintenance requests?
Improvements in data integrity.	Determine how often data must be corrected.	How many corrections were made to data in the last month?

Although the list of questions in Illustration 3.32 is not exhaustive, the questions are a starting point for conducting analyses to address them.

APPLYING CRITICAL THINKING 3.1: Understand AIS Stakeholders

Understanding the stakeholders’ motivations helps accurately identify the objectives of analyses. Stakeholders in AIS data analysis projects can be internal or external to the organization (**Stakeholders**):

- Internal stakeholders include the chief financial officer, internal auditors, managers, chief information officer, and employees. They want to understand and improve processes, so objectives focus on specific aspects of internal processes and how to improve them.
- External stakeholders may include investors, external auditors, customers, and vendors. These stakeholders are motivated by increased return on investment, so the projects with the greatest potential to offer a return on any investment are usually prioritized.

Auditing

Auditors perform data analyses to verify financial statement information is not materially misstated. Their purpose is providing a professional opinion about whether the financial statements provide reliable, accurate information for the client’s primary stakeholders, which include company owners and investors and creditors in the capital markets and financial institutions.

Illustration 3.33 includes examples of motivations, objectives, and questions in auditing data analytics.

ILLUSTRATION 3.33 Motivations, Objectives, and Questions for Auditing Data Analytics

Motivation	Objective	Question
Evaluate the risk of material misstatement of revenue.	Determine if there are any unusual changes in revenue compared to prior years.	Are there any unusual trends or changes in sales from prior years?
Evaluate the risk of material misstatement in the company's general ledger accounts.	Analyze balances in the company's general ledger to identify unusual changes from prior year.	Are any changes in general ledger balances higher than our specified material amount?
Evaluate the risk of fraudulent payments.	Verify employees are not receiving unauthorized payments.	Are there any vendors in the vendor file with the same address as an employee?
Verify physical inventory and fixed asset counts.	Test for existence of assets.	Do inventory counts match amounts on the balance sheet?

Note that the motivation identifies a specific aspect of the audit. The objective states the goal of the analysis. The third column in the illustration provides an example of a specific question the auditor can address with the analysis. Audit analytics are used to perform risk assessment, substantive analytical procedures, and to perform tests of detail.

APPLYING CRITICAL THINKING 3.2: Acquire Auditing Knowledge and Avoid Biases

Auditors acquire and apply certain types of information when performing audit data analytics (**Knowledge**):

- Client's industry, governance, policies, and procedures.
- Relevant accounting and auditing standards, and SEC regulations.
- Risk assessment and statistical and sampling techniques.

Auditors can be biased when they make assumptions about what they expect to find during an audit engagement. They must be vigilantly skeptical of all information they are given (**Risks**).

Financial Accounting

There are a few typical motivations for financial accountants to perform data analyses:

- Ensuring economic transactions, value changes, and period closing entries have been properly captured by the accounting system, valued in the appropriate account, and in the correct accounting period.
- Predicting future net income and cash flows for top management.
- Identifying, evaluating, and securing alternative capital sources.

Illustration 3.34 shows examples of objectives for these motivations and possible questions to address them.

ILLUSTRATION 3.34 Motivations, Objectives, and Questions for Financial Accounting Data Analytics

Motivation	Objective	Question
Ensure the financial statements reflect all transactions.	Verify transactions are not missing.	Does the sales transaction file agree to the financial statements?
Predict cashflows.	Determine expected cash inflow from sales.	Based on prior year sales and expected growth, what are predicted sales for the next year?
Evaluate alternative capital sources.	Decide whether to issue stock or bonds to raise capital.	What is the expected value of issuing more stock based on current market prices?

Managerial Accounting

Managerial accountants perform data analyses to improve management decision-making and operational performance.

APPLYING CRITICAL THINKING 3.3: Consider Alternative Methods

If the objective of an analysis is to predict future income, then managerial accountants may use regression, trendlines, or a what-if analysis. Always choose the method that can answer the questions most effectively and efficiently (**Alternatives**).

Illustration 3.35 shows some typical motivations, objectives, and questions for the projects managerial accountants work on.

ILLUSTRATION 3.35 Motivations, Objectives, and Questions for Managerial Accounting Data Analytics

Motivation	Objective	Question
Identify and evaluate new market, product, service, and business process opportunities.	Determine whether to expand an existing product line.	What are the projected sales for the new product line?
Efficiently allocate resources for successful operations.	Evaluate the best use of materials for production.	Which products use the most materials?
Maximize revenue generation and cash flows.	Determine the most profitable product mix.	How many units of each product should we produce to optimize contribution margin?
Increase organizational efficiency by managing costs.	Determine potential cost cutting.	Which production line has the highest amount of waste?

Tax Accounting

Tax accountants perform data analytics to enhance the quality of their professional advice and decisions, as well as to meet compliance requirements for their organization or clients. Typical motivations for tax analytics include:

- Performing tax research.
- Designing tax plans.
- Correctly calculating tax liability and completing applicable tax returns.

Illustration 3.36 summarizes examples of the objectives related to these motivations and some specific questions.

ILLUSTRATION 3.36 Motivations, Objectives, and Questions for Tax Accounting Data Analytics

Motivation	Objective	Question
A new tax law has been passed that potentially changes a client's tax liability.	Identify relevant tax law changes.	Based on last year's tax return, how will the change in tax law affect the client's tax liability?
A client requested tax planning for the next three years.	Prepare a tax plan to minimize tax liability.	Based on the client's past tax returns and current business projections, how much tax will they owe next year?
Analyzing a client's tax liability after preparing their tax return.	Identify unusual deductions.	How do the client's deductions compare to their prior year return and to national averages?

Although general motivations in tax analytics may be similar, objectives and questions will be specific to individual clients. For example, tax planning is a common motivation, but a client may have differing objectives for their tax plan, such as saving for retirement or estate planning. The objective of the client will determine the type of analysis.

As this chapter has emphasized, accounting professionals are increasingly performing data analysis to provide value to their organizations and clients. Wherever your career takes you, thinking critically about what is motivating a project will help you ask the best objective-based data questions to achieve your goals. These are the first two steps of successful data analysis projects across professional practice areas.

Match at least one professional practice area's acronym to the following motivations to perform data analyses.

- | | |
|-----------------------------------|--------------------------|
| a. Accounting information systems | d. Managerial Accounting |
| b. Auditing | e. Tax Accounting |
| c. Financial Accounting | |

Apply It 3.6
Match Motivations
to Professional
Practice Areas

Motivation	Practice Area
Performing tax research.	
Investing in new technologies.	
Performing analytical procedures on variances in account balances and on classes of transactions.	
Ensuring all economic transactions have been recorded.	
Investing in business intelligence tools.	
Increasing security of AIS assets, including data.	
Innovations in operational processes, either human or computer, to increase process effectiveness.	
Designing defensible tax plans.	
Learning if all adjusting entries at the end of the period have been recorded.	
Improving system performance, such as processing time and availability.	
Reviewing and evaluating the internal control system.	
Increasing security of AIS assets, including data.	
Computing tax liability and completing tax forms.	
Performing physical inventory and fixed asset counts at the fiscal year end.	
Testing documentation to determine if account balances are supported.	

SOLUTION

Motivation	Practice Area
Performing tax research.	Tax accounting
Investing in new technologies.	AIS
Performing analytical procedures on variances in account balances and on classes of transactions.	Auditing
Ensuring all economic transactions have been recorded.	Financial accounting
Investing in business intelligence tools.	AIS
Increasing security of AIS assets, including data.	AIS
Innovations in operational processes, either human or computer, to increase process effectiveness.	Managerial accounting
Designing defensible tax plans.	Tax accounting
Learning if all adjusting entries at the end of the period have been recorded.	Financial accounting
Improving system performance, such as processing time and availability.	AIS
Reviewing and evaluating the internal control system.	Auditing
Increasing security of AIS assets, including data.	AIS
Computing tax liability and completing tax forms.	Tax accounting
Performing physical inventory and fixed asset counts at the fiscal year end.	Financial accounting
Testing documentation to determine if account balances are supported.	Auditing

Chapter Review and Practice

Learning Objectives Review

1 Summarize the relationship between motivations, objectives, and data analysis questions.

Accountants generally perform data analyses for four reasons:

- Opportunities within their organizations and in the marketplace.
- Professional issues and requirements from regulations and laws.
- Solving problems within their organizations or for clients.
- Process and performance evaluation of their own work, or work performed within their organizations.

First, identify the objective, which is the goal of the data analysis. Then, develop questions designed to achieve the project's goal. Data analysis questions can be evaluated for clarity, conciseness, and measurability:

- Will the answer address the objective of the analysis?
- Does the question address a single topic? If not, break it down into sub-questions.
- Is the question measurable?
- Are the data needed to answer it available?

If the answer to any of these questions is no, then either revise the question or drop it.

The six elements of critical thinking can help us think about what is motivating data analyses and how to develop objectives and questions:

- Understanding the stakeholders and their perspectives helps identify the project's motivation and develop specific objectives and questions.
- Critically thinking through the purpose of data analyses supports moving from objective to specific questions.
- Considering several alternative methods of analyses increases the likelihood of selecting the best option for the purpose.
- Being aware of potential risks makes it possible to mitigate them when evaluating motivation, determining objectives, and developing questions.
- Considering what knowledge is necessary to successfully complete the project involves either confirming we already have that knowledge or doing additional research.
- Understanding why we are performing data analyses lets us leverage the experience to future data analytics contexts and tasks. Considering past experiences can speed up the process of identifying objectives and developing questions.

2 Demonstrate how to develop descriptive questions.

Descriptive questions focus on understanding what is currently happening or what has happened in the past. They are developed by identifying the objective of the analysis and then formulating

sub-questions that narrow the focus. The next step is identifying the data and the method of analysis to answer those questions. Common descriptive methods:

- Frequency measures help understand categories of data.
- Measures of location (mean, median, mode) reveal average observations in a data set.
- Measures of dispersion (minimum, maximum, range, variance, and standard deviation) show how much variance there is among the observations in the data set.
- Measures of percentage change show percentage increases and decreases as compared to prior periods and the percent of total.

3 Demonstrate how to develop diagnostic questions.

Diagnostic questions build on what was learned in descriptive analysis and explore the data to find the cause of the outcome. Diagnostic analytics determine why an outcome occurred by looking for anomalies, correlations, patterns, or trends. Common diagnostic analyses:

- Anomaly detection: Scatterplots, bar charts.
- Correlation (linear): Scatterplots, correlation.
- Pattern detection: Line graphs, bar charts.
- Trend analysis: Bar charts, line charts, trendlines.

4 Demonstrate how to develop predictive questions.

Predictive analytics use past and present data to forecast and create models so businesses can make predictions about the future. The increased availability of data and software tools to perform predictive analytics means it is now used in all areas of accounting. To determine a predictive question, ask, "What do I want to do with the answer?" Common analyses used to answer predictive questions:

- Trendlines: Show the underlying functional relationships of the data.
- Regression analysis: Linear regression builds mathematical and statistical models to explain the relationship between a dependent variable and one or more independent variables.

5 Demonstrate how to develop prescriptive questions.

Prescriptive objectives build on predictions about the future and descriptions of the present to determine the best course of action. The objectives for prescriptive analytics focus on what should happen. There are two common methods of prescriptive analysis:

- Optimization is the process of selecting values of variables that *minimize* or *maximize* some quantity of interest. Optimization modeling helps managers allocate resources more efficiently and make cost/profit decisions.

- A spreadsheet model that evaluates how changes and specific combinations of model inputs and assumptions is called a what-if analysis. A what-if analysis is an easy way to change values in a spreadsheet and recalculate the outputs.

6 Describe motivations and objectives for data analytics in professional practice.

The primary professional accounting practice areas have both common and unique motivations to perform data analyses:

- Accounting information systems professionals have governance, strategic, operational, and compliance motivations.
- Auditors have professional compliance, quality, firm, and market motivations.
- Financial accountants have both internal and external stakeholder motivations.
- Managerial accountants have internal governance, strategic, and operational motivations.
- Tax accountants have professional compliance, client, firm, and market motivations.

Key Terms Review

Coefficient of determination 3-21
Constraints 3-25
Correlation coefficient 3-21
Decision variables 3-25
Dependent variable 3-20
Descriptive questions 3-8
Diagnostic questions 3-13

Functional relationship 3-17
Independent variables 3-20
Linear function 3-17
Linear optimization 3-25
Linear regression 3-19
Motivation 3-2
Objective 3-3

Objective function 3-25
Optimization 3-24
Predictive questions 3-16
Prescriptive questions 3-24
Regression statistics 3-20
Variable 3-20
What-if analysis 3-28

How To Walk-Throughs



HOW TO 3.1 Create a Highlight Table in Tableau

The table shown in Illustration 3.10 was created in Tableau version 2021.1.2. Create it yourself by following these steps.

What You Need: **Data** The How To 3.1 data file.

STEP 1: Open a new worksheet and drag the **Sold Date** to **Columns** and **Location** to **Rows**. Drag **Gross Sales** to the open area under years, or to the Text symbol under **Marks** ([Illustration 3.37](#)).

ILLUSTRATION 3.37 Tableau Worksheet

Pages

Filters

Marks

T Automatic

Color

Size

T

Text

Detail

Tooltip

T

SUM(Gross Sale)

iii Columns

Year(Sold Date)

☰ Rows

Model

Sheet 22

		Sold Date		
Location	2023	2024	2025	
Boston	686,611	1,211,197	1,433,578	
Charlotte	618,987	1,254,221	1,653,506	
Chicago	711,852	1,239,689	1,624,719	
Dallas	679,008	1,417,584	1,113,397	
Miami	582,623	1,410,659	1,606,576	
Phoenix	640,681	1,252,712	1,376,503	
Salt Lake City	544,630	1,205,880	1,593,184	
Seattle	460,424	1,514,392	977,384	

STEP 2: Create a calculation that will calculate the change in sales from 2024 to 2025. Because data from 2023 is unnecessary and we are only interested in the Celeritas model, create two filters. First, drag **Model** to the **Filters** box. This opens an input box ([Illustration 3.38](#)).

Filter [Model]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☐ Captain
☐ Celeritas
☐ Kicks
☐ Lazer

All None ☐ Exclude

Summary

Field: [Model]
 Selection: Selected 0 of 4 values
 Wildcard: All
 Condition: None
 Limit: None

Reset OK Cancel Apply

ILLUSTRATION 3.38 Filter Input Box

Check the box for Celeritas and click **OK**. Next, drag **Sold Date** to **Filter** and select **Years** from the input box ([Illustration 3.39](#)).

ILLUSTRATION 3.39 Filter Field Choices

Filter Field [Sold Date]

How do you want to filter on [Sold Date]?

☒ Relative Date
☒ **Range of Dates**

Years
 # Quarters
 # Months
 # Days
 # Week numbers
 # Weekdays
 # Month / Year
 # Month / Day / Year
☐ Individual Dates

Count
 # Count (Distinct)
☐ Minimum
☐ Maximum

☐ Attribute

Next > Cancel

Click **Next** to see the following input box choices ([Illustration 3.40](#)).

ILLUSTRATION 3.40 Filter Choices

Filter [Year of Sold Date]

General

Condition

Top

☒ Select from list

☐ Custom value list

☐ Use all

Enter search text

☐ 2023

☒ 2024

☒ 2025

All

None

☐ Exclude

Summary

Field: [Year of Sold Date]

Selection: Selected 2 of 3 values

Wildcard: All

Condition: None

Limit: None

☐ Filter to latest date value when workbook is opened

Reset

OK

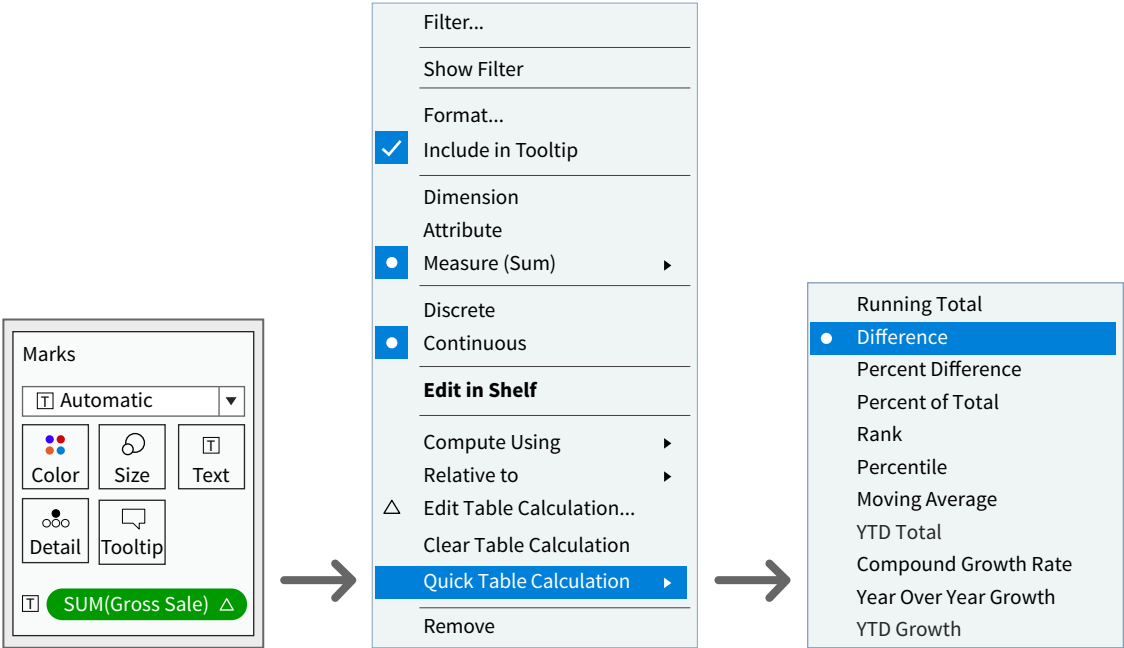
Cancel

Apply

Select 2024 and 2025 and click **OK**.

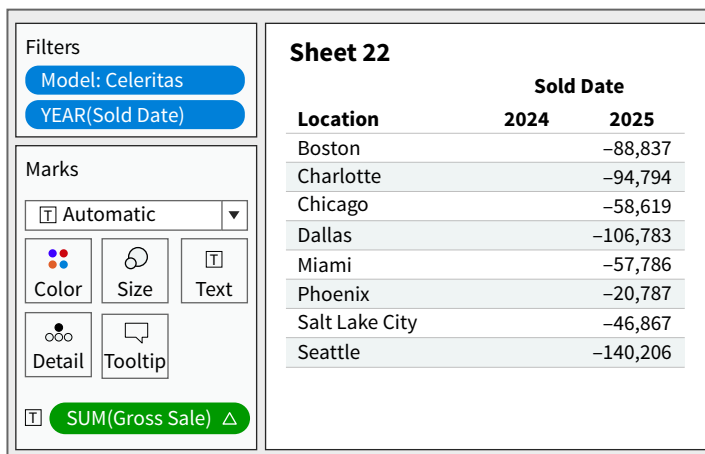
STEP 3: Create the calculation. Clicking the down arrow in the green pill for **Sum(Gross Sales)** in the **Marks** area generates several options ([Illustration 3.41](#)).

ILLUSTRATION 3.41 Creating a Quick Table Calculation



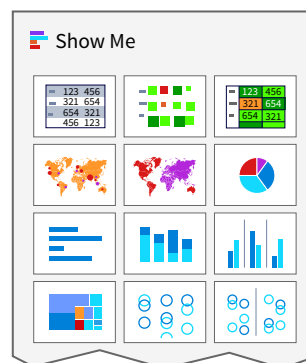
Select the **Quick Table Calculation** option, then select **Difference**. This will change the columns in the table to differences ([Illustration 3.42](#)).

ILLUSTRATION 3.42 Quick Table Calculation Results



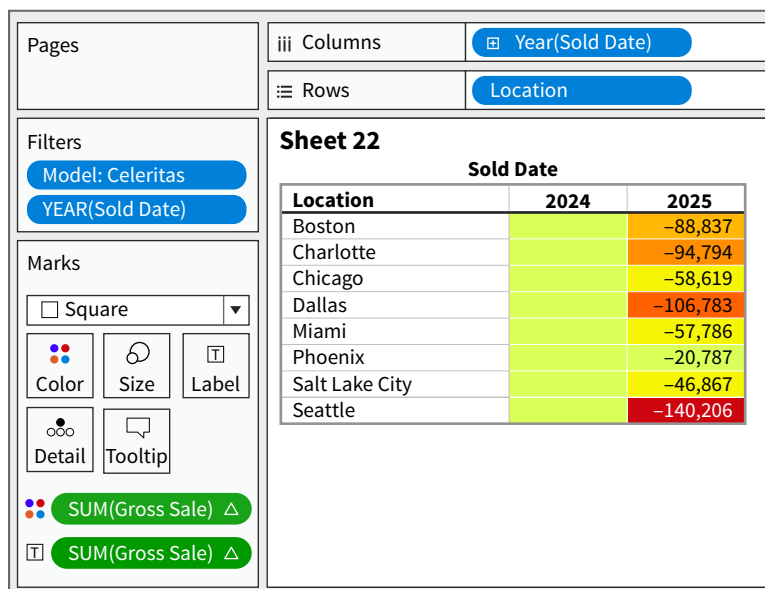
STEP 4: To create the highlight table in which darker colors represent larger numbers, choose that chart type from the **Show Me** options in the top right corner ([Illustration 3.43](#)).

ILLUSTRATION 3.43 Choosing a Highlight Table



Now the table looks like the table in [Illustration 3.44](#).

ILLUSTRATION 3.44 Highlight Table Results



STEP 5: To only show the column for 2025, right click on the 2024 in the column and choose **Hide** ([Illustration 3.45](#)).

ILLUSTRATION 3.45 Hide Column Feature

Sheet 22

Sold Date	
Location	2024 2025
Boston	✓ Keep only
Charlotte	✗ Exclude
Chicago	Hide
Dallas	
Miami	+ Drill down
Phoenix	Format...
Salt Lake City	Rotate Label
Seattle	✓ Show Header
	Show Missing Values

Click on **Hide** and *not* **Exclude**. (Selecting **Exclude** means that the calculation will no longer work because the 2024 data would be removed.)

STEP 6: Order the locations by the largest loss to the smallest loss. To do that, click 2025 at the top of the column. The box will appear as shown in [Illustration 3.46](#).

ILLUSTRATION 3.46 Sorting Data from Smallest to Largest

Sold Date

Location	2025
Boston	-88,337
Charlotte	-94,794
Chicago	-58,619
Dallas	-106,783
Miami	-57,786
Phoenix	-20,787
Salt Lake City	-46,867
Seattle	-140,206

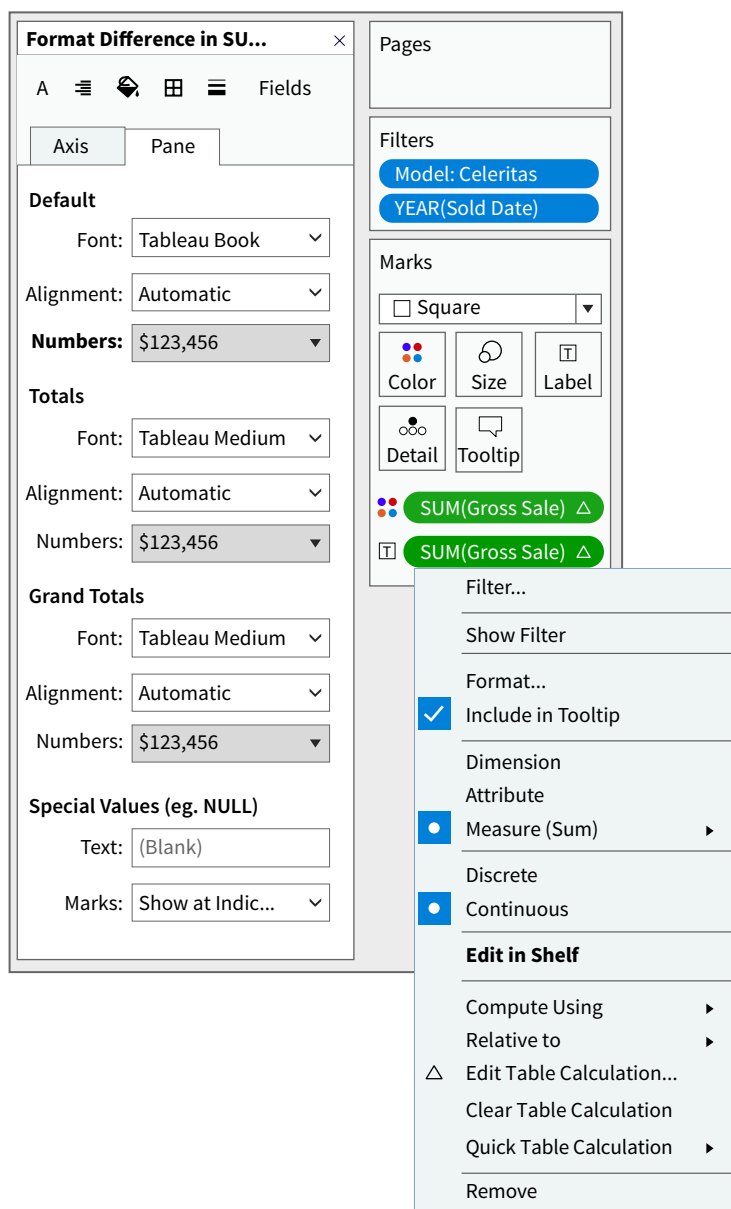
8 items selected Difference in SUM(Gross Sales): -614,679

Click the sort icon that shows the smallest bar on the top. Data will then sort from the smallest number to the largest ([Illustration 3.47](#)).

ILLUSTRATION 3.47 Sorted Highlight Table

Sold Date	
Location	2025
Seattle	-140,206
Dallas	-106,783
Charlotte	-94,794
Boston	-88,837
Chicago	-58,619
Miami	-57,786
Salt Lake City	-46,867
Phoenix	-20,787

STEP 7: Format the difference column to whole dollars by clicking the green **Sum(Gross Sale)** pill in the **Marks** area and choosing **Format** ([Illustration 3.48](#)).

ILLUSTRATION 3.48 Format Options

Here, the default format can be changed to currency.

STEP 8: Create a title for the visualization. Tableau will name the visualization by the sheet number by default. To change that, right click in the sheet title area and then choose **Edit Title** (Illustration 3.49).

ILLUSTRATION 3.49 Editing a Worksheet Title

Sheet 22	
Location	\$
Seattle	\$
Dallas	\$
Charlotte	\$
Boston	\$
Chicago	\$ (-58,619)
Miami	\$ (-57,786)

Enter a title in the box that appears ([Illustration 3.50](#)).

ILLUSTRATION 3.50 Editing a Worksheet Title

Sheet 22

Location	Sold Date 2025
Seattle	\$ (140,206)
Dallas	\$ (106,783)
Charlotte	\$ (94,794)
Boston	\$ (88,837)
Chicago	\$ (58,619)
Miami	\$ (57,786)
Salt Lake City	\$ (46,867)
Phoenix	\$ (20,787)

Edit Title

Tableau Light 15 B I U Insert

Change in Gross Sales from 2024-Celeritas Model

Reset OK Cancel Apply



HOW TO 3.2 Perform a Regression in Microsoft Excel

Illustration 3.18 is regression performed in Excel. You can create it by following these steps.

What You Need: **Data** The How To 3.2 data file.

STEP 1: The Data Analysis tool in the **Data** ribbon has an option for regression. [Illustration 3.51](#) shows the dialog box that opens when **Data Analysis** is selected.

ILLUSTRATION 3.51 Excel Data Analysis Tool - Regression

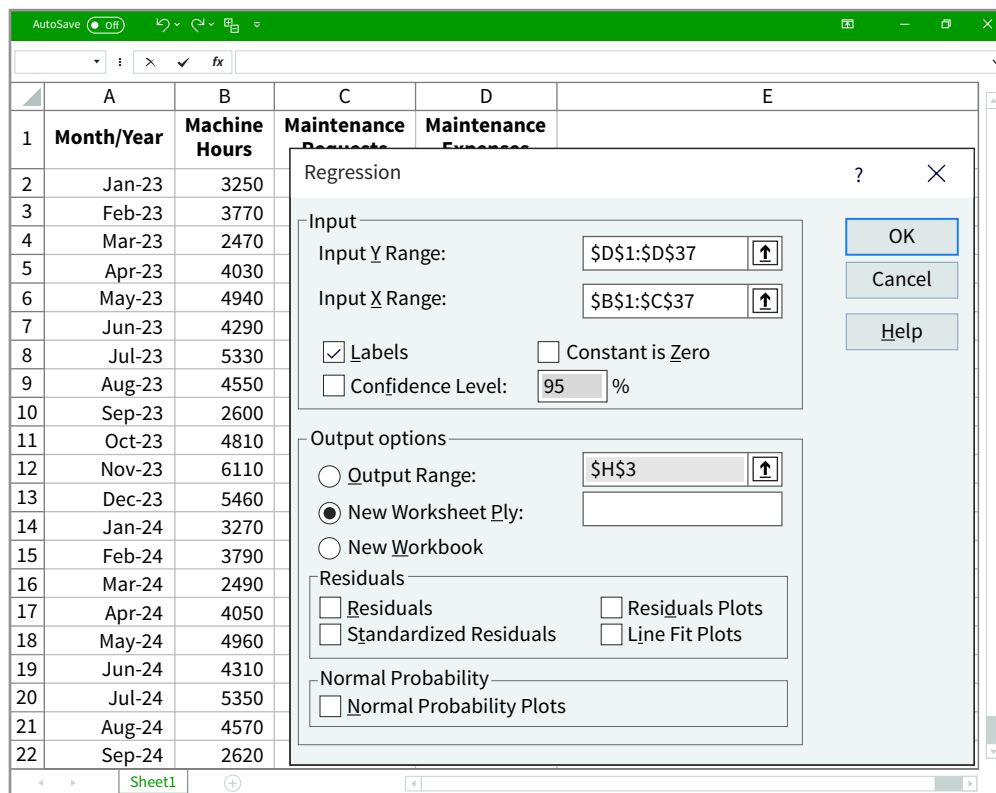
Data Analysis

Analysis Tools

F-Test Two-Sample for Variances
Fourier Analysis
Histogram
Moving Average
Random Number Generation
Rank and Percentile
Regression
Sampling
t-Test: Paired Two Sample for Means
t-Test: Two-Sample Assuming Equal Variances

OK
Cancel
Help

STEP 2: Click on **Regression**, then **OK**. A Regression dialog box will appear ([Illustration 3.52](#)).

ILLUSTRATION 3.52 Regression Dialog Box

STEP 3: The Input Y Range is for the dependent variable. The dependent variable is what will be predicted. In this example, it is column D, Maintenance Expenses (\$D\$1:\$D\$37). The Input X Range is for the independent variables. In this example, there are two independent variables—Machine Hours and Maintenance Requests (\$B\$1:\$C\$37).

Select **Labels** if the first row of the data is the column title (row 1). In this example, the output will be in a new worksheet. Finally, **Residuals**, **Standardized Residuals**, **Residual Plots**, and **Normal Probability Plots** are selected. These residuals options will be used to evaluate the regression assumptions.

STEP 4: Click **OK** for the regression results ([Illustration 3.53](#)).

ILLUSTRATION 3.53 Regression Output

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.978687872				
R Square	0.95782995				
Adjusted R Square	0.955274189				
Standard Error	1283.821342				
Observations	36				
ANOVA					
	df	SS	MS	F	Significance F
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			
	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	5252.858452	897.8427419	5.850533	1.5E-06	3426.183659
Machine Hours	3.569202976	0.43769194	8.154601	2.05E-09	2.678712028
Maintenance Request	759.8366754	168.7763028	4.502034	7.94E-05	416.4587054

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Motivation in data analytics in accounting is
 - a. the reason why the analysis is being performed.
 - b. the interpretation of results of analysis.
 - c. based only on factors outside analysis.
 - d. based only on whether your career will benefit from performing the analysis.
2. **(LO 1)** One source of motivation for data analyses in accounting relates to opportunity. Which of the following is an example of the motivation source of opportunity?
 - a. Desiring future success in your career path.
 - b. Considering whether to expand production of a newly launched product.
 - c. Ensuring compliance with the newest FASB pronouncement.
 - d. Considering why the company's sales volume decreased over a period of time.
3. **(LO 1)** Accountants' motivations to perform data analyses can originate from
 - a. opportunities in the marketplace or organization.
 - b. changes in laws, regulations, and professional rules.
 - c. indications that a problem needs to be solved.
 - d. needing to perform evaluation on processes or performance.
 - e. Any of these may trigger the motivation to perform data analyses.
4. **(LO 1)** Critically thinking through the motivation to perform data analyses includes considering
 - a. stakeholders, purpose, alternatives, risks, knowledge, and self-reflection.
 - b. stakeholders and purpose.
 - c. stakeholders, purpose, risk management, and alternatives.
 - d. stakeholders, acquiring knowledge, alternatives, and self-reflection.
 - e. stakeholders, risk management, alternatives, and self-reflection.
5. **(LO 1)** Analyses of the stakeholders and their perspectives helps to critically think through motivation to perform data analyses because
 - a. it reminds you who pays your salary.
 - b. some may be external to your organization.
 - c. considering the perspectives of stakeholders helps identify the objective and develop good questions.
 - d. internal stakeholders should never be considered.
 - e. All of these are reasons for considering stakeholders in your critical thinking.
6. **(LO 1)** Critically considering the purpose for performing data analyses
 - a. keeps us focused on the reason why something is done or created.
 - b. provides free data for analyses.
 - c. reveals the stakeholders' perspectives.
 - d. helps avoid biases and assumptions.
 - e. None of these list the critical thinking benefits of considering the purpose of data analyses.
7. **(LO 1)** Performing the most appropriate data analyses is more likely if you critically think through issues by
 - a. using bar charts from last year's operational results.
 - b. starting with the results you want to achieve.
 - c. paying others to help you.
 - d. acquiring and applying new knowledge.
 - e. None of these help you perform better data analyses.
8. **(LO 1)** Critically thinking through the risks to your perspective as you determine your motivation to perform data analyses helps you minimize
 - a. errors from biases.
 - b. errors from lack of experience.
 - c. errors from your assumptions.
 - d. faulty conclusions from errors or omissions in your data.
 - e. All of these are benefits of considering the risks to your data analyses choices.
9. **(LO 1)** Considering several alternative methods of analysis helps you
 - a. go with the first alternative that comes to mind.
 - b. select the option that is best for the project's purpose.
 - c. pick the alternative with the highest incremental cost.
 - d. pick the alternative with the lowest likelihood of informative results.
10. **(LO 2)** If the objective of data analysis is to understand what is currently happening or has happened in the past, then what type of question are you asking?

a. Descriptive	c. Predictive
b. Diagnostic	d. Prescriptive
11. **(LO 2)** Once you have articulated a descriptive question, what type of analysis is necessary to understand the dispersion of the data?

a. Median	c. Standard deviation
b. Frequency	d. Percent change
12. **(LO 2)** Assume you plan to perform descriptive analysis of your entity's maintenance expenses. Which of the following analyses would help understand categories of maintenance expenses?
 - a. Create a frequency distribution table based on maintenance expense categories.
 - b. Calculate the mean and median maintenance expense for the period.
 - c. Build a regression model using labor hours as a predictor of maintenance expenses.
 - d. Calculate the percentage change in maintenance expenses since prior year.
 - e. None of these are considered descriptive analysis used to understand categories of maintenance expenses.

13. (LO 3) What type of question is “What is driving the increase in labor expenses?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

14. (LO 3) The sub-question “Are there any unusual sales transactions in 2023?” is best associated with which of the following objectives of data analysis?

- a. What is the average dollar amount of sales in 2023?
- b. What is the best predictor of increased sales for future periods?
- c. Why are sales decreasing during 2023?
- d. What are the minimum and maximum dollar amount of sales during 2023?

15. (LO 4) What type of question is “How much will revenue be next year if we increase sales by 10%?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

16. (LO 4) Which of the following is a question asked in predictive analysis?

- a. Are there any unusual changes in total revenue from 2021 to 2024?
- b. What is the average sales amount for online customers each year?
- c. How will revenue be affected if we have a 5% decrease in variable costs?
- d. How much revenue is categorized as business-to-business sales?

17. (LO 5) What type of question is “How many units of product should we buy each month to minimize inventory holding costs but still meet demand?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

18. (LO 5) In an optimization model, what is being determined?

- a. Constraints
- b. Decision variables
- c. Objective function
- d. How much to produce

19. (LO 6) AIS accountants are most likely to be motivated to perform data analyses for which of the following opportunities?

- a. New technologies that can provide cost savings for operational processes.

b. New customers in the marketplace.

c. Turnover in the regulatory body leadership.

d. Changes to the footnote disclosures required by generally accepted accounting rules.

e. Changes in tax rules.

20. (LO 6) Auditors’ motivation to perform data analyses would be high for which of the following situations?

a. Evaluating the effectiveness of internal controls in their client’s accounting system.

b. Evaluating the completeness of an account balance on their client’s balance sheet or income statement.

c. Considering the risk of material misstatement in engagement planning.

d. Aggregating and analyzing key variables in a smart contract for evaluation regarding whether a client recorded their lease liabilities in accordance with GAAP.

e. All these are examples of situations where auditors are motivated to perform data analyses.

21. (LO 6) Financial accountants would be most likely motivated to perform data analyses for which of the following?

a. Changes to the management team.

b. New competitors in the marketplace.

c. Identifying and supporting assumptions and estimates for end of period adjusting entries.

d. Changes to the tax rates applicable to their organization.

e. None of these would motivate data analyses by financial accountants.

22. (LO 6) Managerial accountants would most likely be motivated to perform data analyses for which of the following?

a. Changes to the tax code.

b. Changes to the financial statement footnote disclosure rules.

c. Changes to reporting rules by the Securities and Exchange Commission.

d. Changes to their organization’s strategies, performance, and resource costs.

e. None of these would motivate data analyses by managerial accountants.

Review Questions

1. (LO 1) Discuss the following motivations for data analyses in accounting by providing an example for each:

1. Opportunities
2. Professional issues and requirements
3. Problem solving
4. Process and performance assessment

2. (LO 1, 6) Compare and contrast the professional considerations and requirements that motivate financial and managerial accountants to perform data analyses.

3. (LO 2, 4) Assume you are a financial analyst at a non-public entity. Your supervisor has asked you to calculate and analyze debt solvency ratios over the last five years. You note that there are several stakeholders associated with this analysis, including your manager

and the bank that holds your entity’s debt. Provide an example of how these two stakeholders may be impacted differently by the results of your data analyses.

4. (LO 1, 4) As an auditor assigned to a public company audit in the manufacturing industry, you have been asked to analyze the company’s inventory to understand the types of products in inventory and the mix of raw materials, work-in-process, and finished goods. You receive the inventory on hand file from the information technology department. Identify the knowledge you might need to acquire or apply to this analysis that may contribute to your motivation to perform it.

5. (LO 2) Assume you are performing a descriptive analysis of the employee expense reimbursement data for your company’s current fiscal year compared to the prior fiscal year. Give an example of an initial question related to this analysis.

6. (LO 2) As an auditor working in the internal audit group, you are asked to perform a descriptive analysis of the employee login attempts during business hours and non-business hours. Your objective is to understand whether the quantity of failed login attempts has increased or decreased. Your initial question is “Have login attempts increased?” Provide an example of a sub-question that would address that objective and initial question for descriptive analysis.
7. (LO 3) Explain how accountants can use diagnostic analytics to determine why an outcome occurred.
8. (LO 3) Your company controller asked you to perform diagnostic analysis on the company’s product sales to understand why sales increased compared to the prior year. Your initial question is “What is driving sales volume increase?” Identify two sub-questions to consider as you plan your analysis.
9. (LO 2, 4) Describe the difference between predictive objectives and descriptive objectives in accounting analytics.
10. (LO 4) Define the terms dependent variable and independent variable. How are these variables used in predictive analytics?
11. (LO 5) Define and explain optimization and what-if analyses. Why are these two types of analyses considered prescriptive analytics?
12. (LO 5) Assume you are a financial analyst working for a manufacturing company. Your company manufactures and sells a variety of

component parts for high-end remote-controlled cars. Your product mix includes three popular products:

- Metal Gear
- High-Torque 330
- High-Torque 400

Your controller has asked you to identify the most profitable product mix. Give an example of an initial question and sub-question that may drive your analysis plan.

13. (LO 6) As an accounting information systems accountant, identify the primary stakeholders you prioritize when performing data analyses.
14. (LO 6) What are external auditors’ primary motivations to perform data analyses?
15. (LO 6) Your controller has asked you to prepare adjusting journal entries for the month-end close. Discuss why a financial reporting accountant may engage in data analysis as they prepare adjusting journal entries.
16. (LO 6) Tax accountants often consider alternatives when deciding whether they are motivated to perform data analyses. Discuss some alternatives that could improve their decisions.

Brief Exercises

BE 3.1 (LO 1) Match the following events with the type of motivation for performing data analyses.

- a. Significant cost increases

b. Annual accomplishments
- c. New industry expertise

d. New standard for leases

Motivation	Event
1. Opportunity	
2. Regulation or law	
3. Problem	
4. Process or performance evaluation	

BE 3.2 (LO 1-6) **Accounting Information Systems** Assume you are an accounting information system professional tasked with performing several analyses related to your company’s information security controls. Match the analytics area to the motivation scenario (analytics areas can be used more than once).

- a. Descriptive analysis

b. Diagnostic analysis
- c. Predictive analysis

d. Prescriptive analysis

Motivation to Perform Analyses	Analytics Area
1. Analyze the mean, median, and mode number of failed login attempts after the company changed password requirements to understand if the benefits of enhanced password requirements outweigh the costs.	
2. Analyze the correlation between the quantity of phishing attempts reported and employee attendance at cybersecurity training to determine if training is related to reporting of phishing attempts.	
3. Analyze trends of failed login attempts over time to determine if employees are complying with company policy regarding login credentials.	
4. Perform a linear regression analysis to consider the likelihood of future information security breaches based on the independent variables of dollars spent on cybersecurity training and encryption software.	
5. Perform a what-if analysis to identify the appropriate dollar amount of spending that is necessary to meet the entity’s goals for information security.	

BE 3.3 (LO 1, 6) Match the motivation source for data analytics in professional accounting to the appropriate scenario. Each source may be used once, more than once, or not at all.

- a. Opportunities
b. Process and performance assessment
c. Regulatory changes
d. Problem solving

Scenario	Motivation Source
1. You are a tax accountant working at a large multi-national company. Executive management is trying to decide the best country in which to expand operations. You have been tasked with identifying independent variables and performing regression analysis to predict potential revenue from this expansion.	
2. Your company has several key performance indicators (KPIs) associated with its manufacturing process – especially quality control. You have been tasked with performing an analysis to identify whether quality control benchmarks are being accurately recorded in the information system.	
3. You are an external auditor working on a public company manufacturing client. You have been asked to test the operating effectiveness of internal controls associated with the purchasing groups' compliance with the internal control that all purchase orders above \$10,000 are approved by a purchasing supervisor.	
4. You are a financial analyst working for a company that distributes consumer products to retailers across the U.S. You must find out why sales of your historically most popular product, the electric grill, have been declining.	

BE 3.4 (LO 2, 6) Auditing As an audit senior, you have asked your intern to perform descriptive analysis on the company's maintenance expenses for the period under audit. You remind the intern that they must first identify the purpose of the analysis and then break that down into questions. The intern is concerned and wonders how they would know if they developed a good descriptive question. Fill in the blank to formulate a response to the intern's concern. Each term can be used once, more than once, or not at all.

Word Bank: objective, specific, available, aspirational, accurate, considerable, exclusive, ambiguous

A good question is one that relates to the _____, is _____, measurable, and can be answered with the _____ data.

BE 3.5 (LO 2) Financial Accounting You are a financial accountant working for the city. Your controller would like to better understand vendor payments made by the city. You have downloaded the vendor payment file for fiscal years 2024 and 2025 from the city's database. An excerpt of the data set is provided.

FY	Activity	Department	Check Date	Vendor	Check Total
2024	Landfill Management	Sanitation Svcs	8/22/2024	Smith Temporaries, Inc	\$ 570.24
2025	Service Maintenance Centers	Public Works and Transportation	8/14/2025	Ferrell Gas, LP	\$ 32.22
2023	Water Treatment	Water Utilities	10/30/2023	Custom-Crete Redi-Mix, LLC	\$ 100.00
2023	Inventory	Inventory Purchase	11/21/2023	Installer Sales and Service	\$ 8.76
2024	Fleet Replacement	Equipment and Fleet Management	6/28/2024	Sam Pack's Five Star Ford	\$ 22,155.00
2024	Homeless Housing Services	Housing/Community Services	6/26/2024	Saint Augustine Estates	\$ 614.00
2024	Complete Street	Public Works and Transportation	3/11/2024	HERC Rentals Inc	\$ 4,236.00
2023	Non-Departmental	Office of Budget	12/24/2023	William H Bancroft Jr.	\$ 2,000.34

- What is the objective of the analysis?
- Develop three questions relevant to the objective, and determine what measures you will use to answer the questions.
- What analyses will you use to answer these three questions?

BE 3.6 (LO 2, 6) Managerial Accounting Match the appropriate analysis that can be used to answer descriptive questions or sub-questions. Each analysis choice can be used once, more than once, or not at all.

- Filter the data to only analyze the machine of interest and use the maximum function.
- Use the minimum and maximum functions to identify the smallest number of units produced and the largest number of units produced.
- Filter the data to only analyze the unit of interest and machine of interest and calculate the mean number of units produced for each shift.
- Create a linear equation to estimate the number of machine hours needed to produce a unit.
- Use the optimization model function in Excel to identify the maximum number of units that can be produced given the constraint of time during shift.
- Create a frequency distribution table to categorize product category.

Initial Descriptive Analysis Questions or Sub-Questions	Analysis Choice
1. How many units were produced for each product category in the facility?	
2. What is the average number of units produced by machine #1,065 during first shift, second shift, and third shift?	
3. What is the greatest number of units that machine #1,810 produced?	
4. What is the range of units produced during the period?	

BE 3.7 (LO 3) Auditing You are an internal auditor at a remote-controlled car manufacturer and sales company. You have been asked to perform an analysis to see if there have been unusual purchases from vendors during the period. Your information technology group provided you with a transaction file that contains all purchases made during the period. An excerpt of the file is provided.

Purchase Order No	Purchase Order Date	Vendor ID	Vendor Name	Item Code	Item Description	Item Quantity	Item Price	Total PO Amount	PO Approved?
2001	1/18/2025	13789	ABC Parts	V473	Cylinder Head	150	\$ 56.24	\$ 8,436.00	
2002	1/18/2025	13356	L&M Manufacturing	V302	Camshaft	250	\$ 57.56	\$ 14,390.00	Y
2003	1/18/2025	13780	TGIF Works	V292	Crankshaft	155	\$ 58.45	\$ 9,059.75	
2004	1/18/2025	13852	LP, LLP	K385	Oil Pan	153	\$ 36.24	\$ 5,544.72	
2005	1/18/2025	13513	GAPISH, CO	K395	Oil Valve	157	\$ 36.47	\$ 5,725.79	
2006	1/19/2025	13253	PTO Park, Co	M390	Nickel Metal Hydride Battery	201	\$ 18.46	\$ 3,710.46	
2007	1/19/2025	13850	STY Manufacturing	M392	Lithium Ion Battery	200	\$ 19.50	\$ 3,900.00	
2008	1/19/2025	13861	Bing Blitz, Inc.	Z389	Struts	268	\$ 37.89	\$ 10,154.52	Y

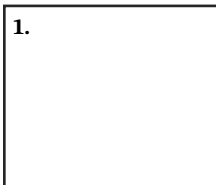
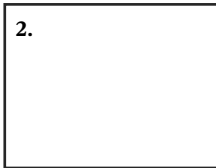
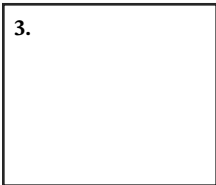
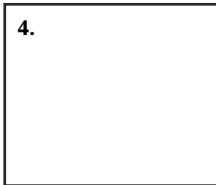
- What is the objective of the analysis?
- Develop three questions that will be relevant to the objective.
- What analyses will you use to answer these three questions?

BE 3.8 (LO 3) Auditing As an auditor for a private entity, your senior has asked you to examine your client's sales returns in the first 30 days after year end. The descriptive analysis revealed sales returns dramatically decreased compared to the prior year. Further, you noted that the largest decrease related to sales returns was in the northwest region. You must perform diagnostic analyses to understand why there was a decrease in sales returns.

Next is an outline of the data analysis objective, initial question, sub-questions, and possible measures. Complete the outline by matching the appropriate statement to the corresponding blank square. Statement choices may be used once, more than once, or not at all.

- What is driving the sales return decrease?
- Are there unusual patterns in the northwest region sales returns for certain product lines?

- c. Create a trend analysis showing returns by date for each product line for the current year compared to the prior year.
- d. Filter data to isolate the northwest region sales returns and examine total sales returns, average sales returns, and sales return quantity.
- e. Create an optimization model to identify the most appropriate number of sales returns given the constraint of sales and quantity manufactured.
- f. Calculate the standard deviation of sales and compare to the standard deviation of employee expenses for the period.
- g. Calculate the overall spending by the company in customer services to determine if sales returns are related to manufacturing quality.

Objective	Initial Question	Sub-questions	Possible Measures
Why did sales returns decrease from prior year?	1. 	Are there anomalies in sales returns for any specific product the company sells? Are there identifiable trends in sales returns for any specific product?	Create a scatterplot comparing quantity returned to sales price for items to identify anomalies in returns. Create a scatterplot comparing sales dollars to sales returns for items to identify anomalies in returns. 2. 
Why is there a large decrease in sales returns in the northwest region?	What factors are driving the decrease in sales returns in the northwest region?	3. 	4. 

BE 3.9 (LO 4) Match the analysis to its description. Each term may be used once, more than once, or not at all.

Analyses:

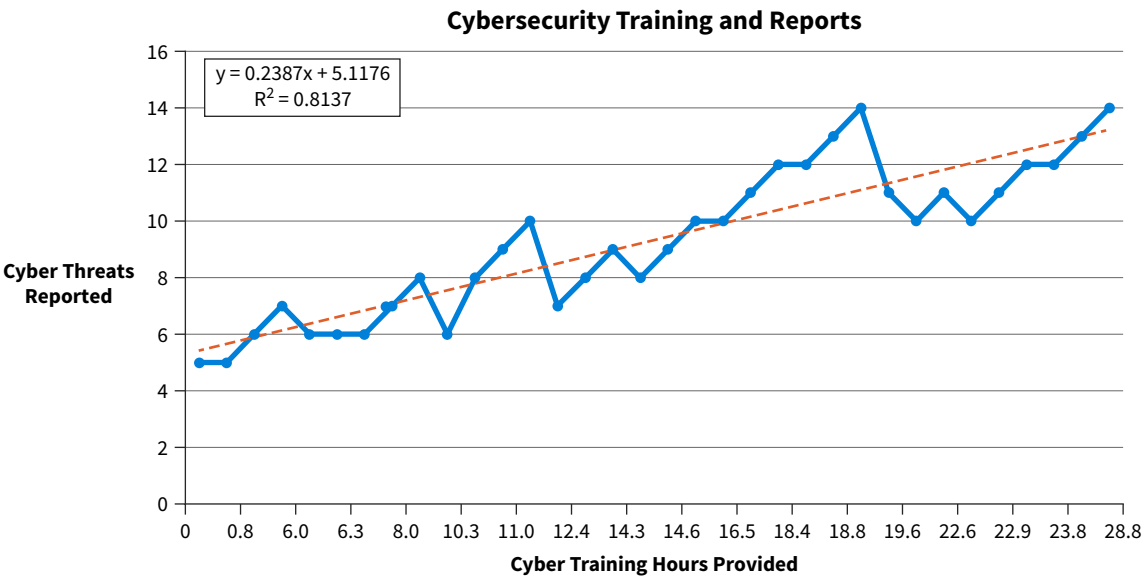
- a. Independent variable
- b. Linear function
- c. Regression statistics
- d. Correlation coefficient
- e. Dependent variable
- f. Adjusted R^2

Description	Analysis
1. The statistical measures used to evaluate a regression model.	
2. The outcome variable in a regression model.	
3. The variable or variables that influence the outcome variable.	
4. This type of relationship shows steady increases or decreases over the range of the independent variable.	
5. This statistic measures the strength of the relationship between the dependent and independent variables.	
6. This statistic explains how well the regression line fits the data.	

BE 3.10 (LO 4) The objective is to engage in predictive analyses. For each of the following initial or sub-question, identify the independent variable and the dependent variable.

- How much will revenue change with a 15% increase in sales volume?
- How much will revenue increase with a 10% increase in production volume?
- How much will maintenance expenses change if we increase production volume by 5%?
- As unemployment increases, how much will sales volume decrease?
- As the number of employees increase, how much will information security costs increase?

BE 3.11 (LO 4, 6) Accounting Information Systems Assume you are an accounting information systems accountant asked to consider the success of your company’s cyber training program. You collected data on the monthly number of cybersecurity training hours provided to employees and the number of cyberthreats reported. You created the following trendline line graph.



- Use the information provided in the line graph and the equation to answer the following questions:
- If the company offered 9 hours of cyber training during a period, what is the expected number of cybersecurity threats reported by employees?
 - If the company offered 35 hours of cyber training during a period, what is the expected number of cybersecurity threats reported by employees?

BE 3.12 (LO 5) Match the appropriate term to each definition. Terms can be used once, more than once, or not at all.

- | | |
|-----------------------|-------------------------|
| a. Constraints | e. Linear regression |
| b. Objective function | f. Dependent variable |
| c. Decision variable | g. Independent variable |
| d. Optimization | |

Definition	Term
1. The process of selecting values of variables that minimize or maximize some quantity of interest.	
2. The unknown values a model seeks to determine.	
3. The mathematical equation that describes the output target that we seek to minimize or maximize.	
4. The limitations, requirements, or other restrictions that must be imposed on any solution.	

BE 3.13 (LO 5, 6) Managerial Accounting As a managerial accountant for a manufacturing company, you are asked to identify the optimal number of units that must be produced in order to maximize the company's contribution margin (CM). Your company manufactures four different types of units:

- Standard widgets
- Blue flying widgets
- Red swimming widgets
- Yellow hopping widgets

To perform prescriptive analysis, your manager has built a spreadsheet for you to execute a Microsoft Excel Solver function.

	Standard	Blue Flying	Red Swimming	Yellow Hopping	Total
Number Produced	1	1	1	1	
CM per unit	17.87	15.20	21.00	19.87	
Total CM	17.87	15.20	21.00	19.87	73.94
Machine Hours Per Unit	6	8	12	11	0
Total Machine Hours	6	8	12	11	37
Constraints					
Machine Hours					50,000
Demand	1,200	1,750	1,670	1,250	

1. What is the decision variable in the model?
2. What are the constraints in the model?
3. What is the objective function?

BE 3.14 (LO 5, 6) Financial Accounting Your private company wants to expand and grow into a new product line; however, that growth requires increased capital and investment in physical assets, people, and materials. You are asked to consider the most appropriate strategy to acquire the needed capital. You volunteered to engage in data analysis to provide insight to the discussion.

1. What is the objective of the analysis?
2. Develop two questions that will be relevant to the analysis.
3. What analyses will you use to answer each question?

BE 3.15 (LO 6) Identify whether the professional has developed a measurable and specific question appropriate for the practice area.

1. A financial statement auditor asks, "What is the expected value of the client issuing stock based on current market prices?"
2. An accounting information systems accountant asks, "How many units should the company produce to optimize the company's contribution margin?"
3. A tax accountant poses the question, "Based on last year's tax return, how will the current year changes in tax law impact the client's tax liability?"
4. A managerial accountant considers "Which products use the highest amount of raw materials in production?"
5. A financial accountant ponders "What dollar amount of accounts receivable are classified in the over 90 days outstanding category?"

BE 3.16 (LO 6) Match each statement to the most appropriate accounting professional practice area. Each accounting professional practice can be used once, more than once, or not at all.

- a. Accounting information systems
b. Financial accounting
c. Managerial accounting
- d. Auditing
e. Tax accounting

Statement	Practice Area
1. Simone and the information technology team are investigating the possibility of investing in new technologies.	
2. Chung would like to improve the company’s information system performance and is evaluating processing time to prepare month-end management reports.	
3. Christine wants to identify unusual sales transactions that may have contributed to an unexpected increase in the client’s revenue compared to last year.	
4. Daniel is currently using the Scenario Manager function in Excel to identify interest rates, payment terms, and length of a loan to evaluate various capital requirement strategies.	
5. Ellis is analyzing a client’s tax liability calculation to identify if there are expenses that are not properly included as deductions.	
6. Pierre is currently examining the relationship between machine hours and maintenance expenses to predict future manufacturing modifications.	
7. Mica is performing a regression analysis to estimate the predicted monthly revenue for a retail client. He will then compare the predicted monthly revenue to the client’s recorded amount.	

Exercises

EX 3.1 (LO 2, 6) Data Financial Accounting Analysis of Measures As the company controller for a yarn manufacturing supply company, you are asked to investigate sales by customer for the month of June 2025. You have critically assessed the objective and questions of your analysis and documented them as follows:

Objective	Initial Question	Sub-questions	Possible Measures
Understand sales by customer for the month of June.	What happened to sales in the month of June?	What was total sales? What was the average sales amount? What was total sales by customer? Which customers had the highest sales?	1. Sum of sales amount (total measure). 2. Average of sales amount (measure of location). 3. Total sales separated by customer. 4. Total sales separated by customer using a highlight table.

Use the available data to perform the analyses proposed in the possible measures section of the analysis plan.

EX 3.2 (LO 2, 6) Data Managerial Accounting Analysis of Measures Your company manufactures high-quality yarn for sale at craft stores and yarn and specialty shops across the country. Specifically, your facility manufactures merino wool yarn in a variety of colors and in two weights: chunky and DK weight. The product manager wants to understand differences in production quantities by product between 2025 and 2024 in the July manufacturing period, and has provided you with production data for select products. Use the spreadsheet to identify the available data and complete the following:

- What is the objective of the analysis?
- What are two initial questions you can ask about the data?
- Create a column chart depicting the production quantity by ProductDescription and Year.
- Calculate and identify several descriptive measures:
 - Calculate the Average ProductionQuantity by ProductDescription in 2024 compared to 2025.
 - Identify the lowest number of units produced by ProductionDescription in 2024 compared to 2025.
 - Identify the highest number of units produced by ProductionDescription in 2024 compared to 2025.

EX 3.3 (LO 2, 6) Data Financial Accounting Tableau Highlight Table Your controller would like to better understand changes in specific vendor payments between 2025 and 2024 that were made by the city you work for. The controller is specifically interested in understanding the payments made to the following vendors:

- 4-Star Hose & Supply
- Aecom Technical Services, Inc.
- WRG, LLC
- Winston Water Cooler Ltd.
- Zoetis Inc.

You have downloaded the vendor payment file and uploaded it into Tableau for analysis. Create a highlight table in Tableau to identify changes in payments to each of these five vendors between 2024 and 2025.

EX 3.4 (LO 2, 6) Data Accounting Information Systems Analysis of Measures As part of your company's new cybersecurity program, each employee must participate in an annual training. The trainings are conducted in a variety of modalities (in-person, online synchronous, and online asynchronous). You and your team have tracked the training expenses each month as well as the number of cybersecurity threats reported by employees. Employees can report cybersecurity threats such as phishing attempts, or IT specialists can report threats through examination of the monthly login report. You have been asked to investigate cybersecurity training costs over the period from January 2023 through December 2025.

Review and complete the chart by identifying sub-questions and possible measures used to address your identified sub-questions.

Objective	Initial Question	Sub-questions	Possible Measures
Understand cybersecurity training expenses.	What is the total amount of cybersecurity training expense for 2023, 2024 and 2025?	1.	2.
	What is the percentage change of cybersecurity training expenses between 2023 and 2024 and between 2024 and 2025?	3.	4.

5. Then, conduct your descriptive analyses using the provided data file.

EX 3.5 (LO 3, 6) Data Auditing Diagnostic Measures Assume you are an audit staff assigned to audit the financial statements of a private entity distribution client for the year ended December 31, 2025. The client does not manufacture items, but acquires finished products and sells to a variety of retailers. The senior on the engagement team asked you to perform analytical procedures on sales returns. Your descriptive analysis revealed that sales returns increased as a percentage of sales. Use the spreadsheet to perform the following:

- Calculate total sales dollars, returns dollars, sales quantity, and returns quantity for each year 2024 and 2025.
- Calculate the percentage change in sales dollars, returns dollars, sales quantity, and returns quantity between 2024 and 2025.
- Create a scatterplot whereby sales dollars are on the x-axis and the sales returns are on the y-axis.
- Which months in 2025 have sales returns that you may consider as anomalies?

EX 3.6 (LO 3, 6) Data Managerial Accounting Trend Analysis with Line Graphs You are an intern for a manufacturing company, and your company controller has learned you have completed a data analytics course with an emphasis in accounting. The controller has performed basic descriptive analysis of the company's sales for the month of June 2025 but wants to understand why sales increased 5.14%. Use the data set to complete the following:

- What is the initial question for this analysis?
- Identify two sub-questions for the analysis.
- Identify potential measures that you may calculate to answer these two sub-questions.
- Create a line graph showing the trend in June sales in 2024 compared to 2025.
- Create a column bar chart showing sales to each customer during June 2024 compared to June 2025.

EX 3.7 (LO 3, 6) Financial Accounting Trend and Correlation Analysis The corporate controller asked the team to analyze accounts receivable and the aging of accounts receivable compared to prior year. Your descriptive analysis indicates accounts receivable increased over prior year, while sales have remained steady. Since you are the intern on the job, your colleagues have asked you to consider the diagnostic analysis questions that you must ask to plan your analysis. You were provided with the data dictionary.

Data Field	Description
SalesOrderNumber	Uniquely assigned sales order number.
CustomerID	Uniquely assigned customer number.
CustomerName	A customers name.
CustomerCredit	Customer’s credit limit.
CustomerAddress	U.S Postal mailing address and street name.
CustomerCity	Mailing address city.
CustomerState	Mailing address state.
CustomerZipcode	Mailing address zip.
CustomerPayTerms	Payment terms negotiated with the customer.
SalesOrderDate	Sales order date.
SalesOrderID	Uniquely identifiable item number.
SalesOrderDescription	Description of item.
ItemPrice	Sales price of the item per negotiated terms with the customer.
ItemQuantity	Total quantity of items sold to the customer.
InvoiceNumber	Uniquely assigned invoice number.
InvoiceAmount	The total amount of the invoice, including shipping charges and tax.
PaidAmount	The total amount of the payment received from the customer.
PaidDate	The date the invoice was paid.

Complete the diagnostic objective questions in the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Understand why accounts receivable increased compared to the prior year.	1.	2.	3.

EX 3.8 (LO 3, 6) Data Accounting Information Systems Trend and Correlation Analysis You and your team have tracked the number of cybersecurity threats reported by employees. Employees can report cybersecurity threats such as phishing attempts, and IT specialists can report threats through examination of the monthly login report.

Your initial descriptive analysis revealed that in the last three years, 313 cyber incidents were reported. There were 73 reports in 2023, 111 in 2024, and 129 in 2025. Your manager would like to understand why cyber incident reports have increased. You have started to identify initial questions, sub-questions, and possible measures for your analysis in the following chart. Complete the chart.

Objective	Initial Questions	Sub-questions	Possible Measures
Determine why cyber incident reports are increasing.	1.	Are there more cyber incidents reporting during a specific period of time?	Create a line chart of cyber incident reports over time and review for a trend or pattern.
	2.	3.	Perform a correlation between training expenses and cyber incidents reported by using the CORREL function in Excel.

4. Perform the possible analyses.

EX 3.9 (LO 4, 6) Data Accounting Information Systems Trendlines Your controller has asked you to demonstrate the relationship between cybersecurity training expenses and the number of valid cybersecurity threats reported by employees. You have collected monthly data reporting the monthly cybersecurity training expense and the number of valid cybersecurity threats reported by employees. Create a line graph showing the relationship between these two variables. Include the trendline, equation, and R-square of your analysis.

EX 3.10 (LO 5, 6) Data Auditing Sales Returns Analytical Procedures You are the audit intern staffed on the fiscal year ended December 31, 2025 financial statement audit of a private company. Your audit senior is thrilled that you have just completed a data analytics in accounting course and wants you to use your knowledge of predictive analytics to develop an expectation of sales returns based on sales quantity to perform analytical procedures. You have engaged in critical thinking and developed the partial chart.

Complete the chart:

Objective	Initial Question	Sub-question	Possible Measures
Predict sales returns.	How much will sales returns change if we have a 10% increase in sales quantity?	1.	Regression analysis

2. Perform the regression analysis to predict sales return dollars using sales quantity.
 3. Write the regression equation.
 4. What is the R^2 for the regression model?

EX 3.11 (LO 5, 6) Data Financial Accounting Estimate Warranty Expense for Accruals As a financial accountant for a public widget manufacturing company, you are given a data file and asked to calculate and record month-end adjusting journal entries. You must calculate the warranty expense amount for January 2025.

1. Perform a regression to predict warranty expense using sales quantity.
 2. Write the regression equation.
 3. You just performed a query of your company's accounting information system and determined that January 2026 sales quantity is 10,385. Write the journal entry to record warranty expense.

EX 3.12 (LO 5, 6) Data Managerial Accounting Trend Analysis You are a financial analyst supporting the control in a private company that manufactures paper plates, trays, and other disposable paper products. The company is owned by a private equity (PE) investment group. The PE board is interested in cost savings and identifying efficiencies within the company's processes. You collected monthly data on the number of die-cut changes and the monthly maintenance expenses. The goal is to predict the cost of maintenance expenses for the next period.

1. Run a regression analysis using die-cut changes to predict maintenance expenses.
 2. Write the regression equation.
 3. If the company plans to have 650 die-cut changes during the month of January 2024, what is the company's expected maintenance expense?

EX 3.13 (LO 4, 6) Data Auditing Predictive Analytics Kahn & Williams CPA firm are the external auditors for Super Scooters. The audit team is performing substantive procedures on revenue and designed a test that uses predictive analytics to evaluate the reasonableness of Super Scooters reported revenue. The objective is to evaluate recorded revenue for reasonableness. The initial question is: Does reported monthly revenue vary significantly with the predicted monthly revenue from the engagement team's regression model? The team's model is based on the prior three years of monthly data. Further, the team believes that revenue is driven by sales volume and the average price of scooters sold during the month. The team has decided that any difference between predicted revenue and the actual revenue that is greater than the standard error (in absolute dollars), should be investigated further.

The summary output is given here:

PREDICTED REVENUE MODEL

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.997170819							
R Square	0.994349643							
Adjusted R Square	0.994007197							
Standard Error	27302.53042							
Observations	36							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	4,328,953,815,697.83	2,164,476,907,848.92	2,903.67	0.00			
Residual	33	24,599,129,513.81	745,428,167.09					
Total	35	4,353,552,945,211.64						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-597355.3555	79141.28598	-7.547961195	1.10769E-08	-758369.5125	-436341.1985	-758369.5125	-436341.1985
Volume	424.0763009	7.525920213	56.34876386	2.20655E-34	408.7647011	439.3879007	408.7647011	439.3879007
Average Price	1453.623412	208.2682902	6.97957145	5.57457E-08	1029.89839	1877.348434	1029.89839	1877.348434

1. What is the dependent variable in the model (y range)?
2. What are the independent variables in the model (x range)?
3. What is the regression equation for this model?

EX 3.14 (LO 5, 6) Data Managerial Accounting Optimization Your production manager has asked you to identify the quantity of items that should be produced to maximize the company's contribution margin for the period. Your company has four products:

- Standard widgets
- Blue flying widgets
- Red swimming widgets
- Yellow hopping widgets

You have created a spreadsheet outlining the demand, machine hours per unit, contribution margin per unit, and overall machine hours. Use Microsoft Solver to identify the number of units to be produced for each product and the maximum contribution margin.

EX 3.15 (LO 5) Data Managerial Accounting Use Linear Optimization to Determine Product Mix and Profit You and your college roommates open a business designing and crafting dog and cat leashes with your college logo and colors. You have decided to offer leashes in three lengths: 6, 8, and 12 feet.

Further, you have two types of leash: "standard" or "decorative." While the decorative leashes take longer to create, they have a large profit margin per each. However, the standard leashes are in higher demand. You and your two roommates have put a schedule together so that you can fit everything into your week – this includes classes, social events, and crafting time. Since you are enrolled in the data analytics in accounting course, you have volunteered to calculate how many of each type of leash you should produce weekly based on demand, gross profit for each product, and the constraint of work hours.

1. Perform a linear optimization using Excel Solver.
2. How many of each unit should you and your two roommates produce each week?
3. What is the total gross profit you should expect to earn based on the weekly production of dog leashes?
4. Are there any constraints identified in the answer report that are non-binding?

EX 3.16 (LO 4, 5) Data Build the Spreadsheet and Identify Constraints Along with your college roommates, you are opening a business to make decorative hats, T-shirts, and socks with your college colors. You have decided to start with five core products:

- Decorative ball cap
- Decorative “GO Team” t-shirt
- Decorative “Defense Wins Championships” t-shirt
- Decorative socks
- Decorative knee-high socks

As part of your business plan, you want to calculate the total number of each product to create each week. You plan to use Microsoft Solver to identify the maximum number of units to produce given your constraints. You want to maximize your company’s profit given the constraints of work hours per person. The table outlines the gross profit and crafting hours per unit for each item.

Product	Gross Profit per Item	Crafting Hours per Item
Ball cap	5.00	4
GO Team t-shirt	7.50	2.5
Defense Wins Championships t-shirt	7.50	3
Socks	3.30	6
Knee-high socks	4.50	3.5

In addition, you and your roommates have committed to working the following hours each week:

- You: 20
 - Roommate 1: 20
 - Roommate 2: 15
 - Roommate 3: 10
1. Build a linear optimization model spreadsheet.
 2. Calculate the total number of units for each product that should be produced each week.
 3. What is the total gross profit you should expect to earn based on the weekly production?
 4. Are there any constraints identified in the answer report that are non-binding?

EX 3.17 (LO 5, 6) Data Managerial Accounting Linear Optimization Tiana’s Jewel Design, Inc. designs and manufactures mass produced metal jewelry and custom designed silver-plated jewelry. Currently, the owner designs and produces three lines of custom jewelry: necklaces, bracelets, and earrings (decision variables). The contribution margin and the resource requirements for each product follows.

	Necklaces	Bracelets	Earrings
Contribution Margin per unit	\$ 12.00	\$ 8.00	\$ 14.00
Resource Requirements:			
Direct Materials, ounces per unit	22	8	13
Design Hours per unit	4	1	5
Machining Hours per unit	2	1	2

They would like to maximize the contribution margin (objective function). However, the owner has these constraints:

- A contract to produce at least ten pairs of custom earrings a month.
 - They can sell all the custom necklaces and custom earrings that are produced
 - Demand for custom bracelets is only 30 a month.
 - Each month 800 ounces of silver plate can be purchased.
 - Two designers work a total of 120 hours a month.
 - An estimated 80 hours of machining time are available each month.
1. Use Microsoft Excel Solver to create a linear optimization model.
 2. How many of each product should be produced to maximize the contribution margin?
 3. Are any of the constraints binding?
 4. Based on the output from the answer report, what advice would you give the owner?

Professional Application Case: Healthcare Service Company



Kindred Healthcare is a post-acute healthcare service company that operates long-term acute care and rehabilitation services across the United States. Kindred's hospitals offer the same in-depth care that a patient would receive at a traditional hospital, but for an extended recovery period. The data used in this case is based on actual data from the California Health and Human Services Open Data Portal.

Kindred has 13 long-term acute care hospitals in California:

- Baldwin Park
- Brea
- La Mirada
- Los Angeles
- Ontario
- Paramount
- Rancho
- Riverside
- Sacramento
- San Diego
- San Francisco Bay Area
- South Bay
- Westminster

The hospital locations in California are consolidated into financial statements which are audited by external auditors and provided to the board of directors, the company's primary investors, and the financial institutions who have extended the bank loans. The income statement for the prior three years is provided.

	2019 (unaudited)	2018 (audited)	2017 (audited)
Gross Patient Revenue	\$2,802,211,999	\$2,515,881,547	\$2,423,431,540
Deductions from Revenue	\$2,206,888,182	\$1,940,233,238	\$1,869,518,824
Net Patient Revenue	\$ 595,323,817	\$ 575,648,309	\$ 553,912,716
Other Operating Revenue	\$ 2,262,385	\$ 2,367,007	\$ 2,663,316
Operating Expenses	\$ 579,928,100	\$ 523,090,033	\$ 506,521,633
Net Revenue from Operations	\$ 17,658,102	\$ 54,925,283	\$ 50,054,399
Non-Operating Revenue	\$ 1,237,718	\$ 970,497	\$ 763,174
Non-Operating Expense	\$ 110,477	\$ 200,367	\$ 385,724
Income Tax	\$ 86,442	\$ 73,200	\$ 25,176
Net Income	\$ 18,698,901	\$ 55,622,213	\$ 50,406,673

The financial statements are prepared based on annual amounts from each of the California locations. The consolidated entity has provided a data file that contains annual data by location for a variety of data fields.

PAC 3.1 Auditing: Develop Objectives and Questions for an Audit Plan

Data Auditing You are a second-year staff member at a public accounting firm who is assigned to the Kindred Hospitals engagement, which is a new engagement for the firm. You have been asked to perform descriptive, diagnostic, and predictive analytics for your engagement team.

The audit senior provided you with an Excel file that includes gross patient revenue by hospital location from 2013 to 2019. The senior asked you to perform exploratory data analytics and outline an audit plan, which you recognize is similar to outlining objectives, questions, and sub-questions for data analysis. Before you can fully explore the data, you must understand it, so you begin with descriptive statistics.

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross patient revenue (GrossPatientRevenue) for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross patient revenue for the year 2019 compared to 2018. Specifically, GrossPatientRevenue, GrossInpatientRevenue, and GrossOutpatientRevenue data points.	Which locations had changes in revenue from 2018 to 2019?	3.	4.
Predictive Analysis: Develop an expectation for gross patient revenue in 2019 based on the components patient days, out patient visits, and average length of stay.	What factors drive changes in gross revenue?	5.	6.

7. Perform the possible analytics as described for each measure.

PAC 3.2
Managerial Accounting: Evaluate Labor Costs and Productivity

Data **Managerial Accounting** Assume you are a managerial accountant for the consolidated California group of Kindred Hospitals. The director of operations has asked that you analyze the annual data associated with hospital labor. Specifically, the director of operations is interested in the following variables.

Variables	Definition per Data Dictionary
HopsitalFTE	Number of hospital full time employees
NurseFTE	Average number of nursing personnel
ProductiveHours	Total productive hours
NonProductiveHours	Total nonproductive hours
PaidHours	Total paid hours
MedicalStaff	Number of active medical staff

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand labor information for 2018 - 2019.	What happened to labor hours during 2018 and 2019?	1.	2.
Diagnostic Analysis: Determine why variables changed between 2018 and 2019.	Which hospitals have increasing salary expenses from 2018 to 2019?	3.	4.
Predictive Analysis: Identify if there is relationship between total productive hours and gross patient revenue.	Can productive hours predict gross patient revenue?	5.	6.

7. Use the data to calculate the measures.

PAC 3.3 Financial Accounting: Understand Patient Revenue

Data Financial Accounting You are a financial analyst working for the controller's group for the California location of Kindred Hospital group. Your controller has highlighted the fact that gross revenue consists of both inpatient revenue and outpatient revenue. However, only one facility – Kindred Hospital-Rancho – has outpatient revenue. Therefore, the controller wants a full understanding of inpatient revenue by facility for 2019. Specifically, you are asked to understand the dollar amount of inpatient revenue contributed by each facility in 2019, and how inpatient revenue has changed since 2018. Finally, the controller is creating a financial forecast for inpatient revenue for 2020 and wants to understand the utilization data that may contribute to the generation of this forecast. To satisfy your company controller, you are planning your analysis and have started the chart which follows. The controller has also provided you with a data dictionary for the data relevant to this analysis.

Variables	Definition per Data Dictionary
DayTotal	Patient Days Total
OccupancyLicensed	Occupancy rate – Licensed beds
OccupancyAvailable	Occupancy rate – Available beds
AverageLengthStayAll	Average Length of Stay – Including Long Term Care
AverageLengthStayNoLTC	Average Length of Stay – Excluding Long Term Care
GrossInpatientRevenue	Gross Inpatient Revenue

Complete the chart for descriptive, diagnostic, and predictive analytics.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross inpatient revenue for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross inpatient revenue for the year 2019 compared to 2018.	How has revenue changed at the locations?	3.	4.
Predictive Analysis: Identify the utilization measures that are related to in-patient revenue.	Are there any measures correlated to inpatient revenue?	5.	6.

7. Use the data available to calculate the measures.

PAC 3.4 Tax Accounting: Evaluate Non-Profit and For-Profit Tax Return Data

Data Tax Accounting As a tax accountant at Kindred you have been asked to analyze the non-profit and for-profit hospital data. Kindred Hospitals has three hospitals in California that are tax exempt organizations. These three hospitals must be separated from the for-profit hospitals for tax purposes. In addition, you have been asked to do an analysis of a sample of non-profit tax filings and compare those filings to the amounts that Kindred's non-profit hospitals filed for 2019. The variables and data definitions for both files are included in the following tables:

IRS 990 Data:

Variables	Definition per Data Dictionary
Assets	Total asset book value at year end
Income	Net Income
Revenue	Revenue reported on line 12 of Form 990

Kindred 2013 – 2019 Financial Data

Variables	Definition per Data Dictionary
TotalAssets	Total Assets
TotalRevenue	Net patient revenue plus other operating revenue
NetIncome	Net Income

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand assets, revenue, and income information for Kindred's non-profit and for-profit hospitals	Which hospitals have the highest income?	1.	2.
Descriptive Analysis: Understand how IRS 990 data for non-profit hospitals compares to the Kindred non-profit hospitals	Are the Kindred Hospitals comparable to the sample of IRS 990 filings.	3.	4.
Diagnostic Analysis: Identify trends in assets, revenue, and income for Kindred hospitals.	5.	6.	7.
Diagnostic Analysis: Compare Kindred data to IRS 990 data.	8.	9.	10.

11. Use the data to calculate the measures.

Le Grind Continuing Case: Motivation and Specific Questions for Analyzing Gross Profit

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.

